

ARCOS Group

uc3m | Universidad **Carlos III** de Madrid

L3: Fundamentals of assembler programming (2) Computer Structure

Bachelor in Computer Science and Engineering

Bachelor in Applied Mathematics and Computing

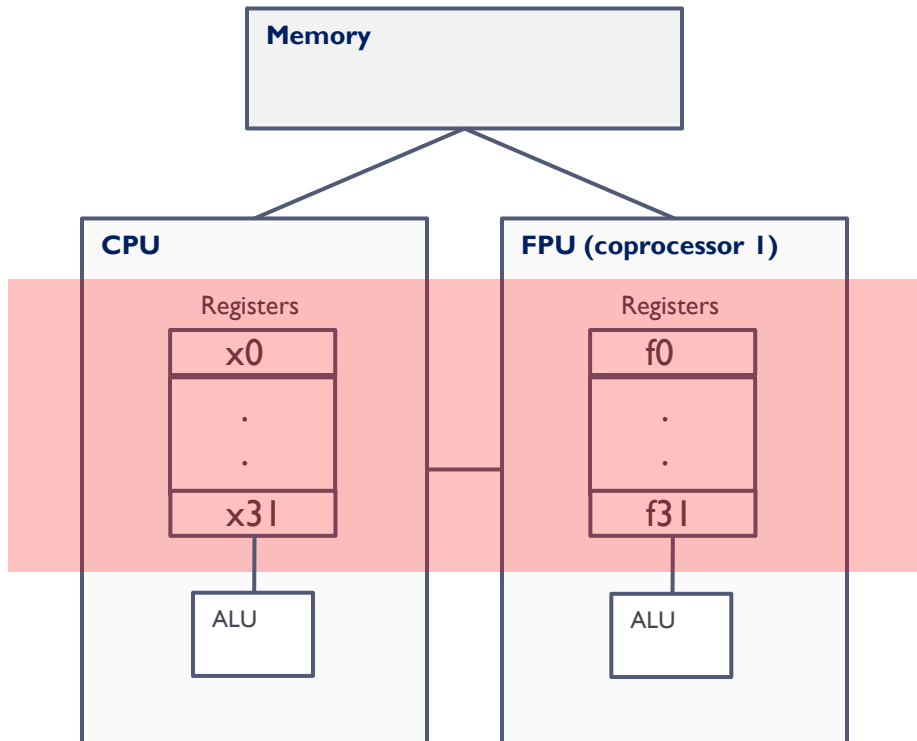
Dual Bachelor in Computer Science and Engineering and Business Administration



Contents

1. Basic concepts on assembly programming
2. RISC-V 32 assembly, memory model and data representation
 - ▶ RISC-V: registers and memory
 - ▶ Assembler directives
 - ▶ System services
 - ▶ Memory access instructions
3. Instruction formats and addressing modes
4. Floating point in assembly
5. Procedure calls and stack convention

RISC-V 32



▶ Registers

- ▶ Inside the processor
- ▶ Instructions work with values in registers
- ▶ 32 registers of 32 bits

▶ Memory

- ▶ Outside the processor
- ▶ Data exchange to/from registers
- ▶ More capacity but longer access time than registers
- ▶ 32-bit addresses
 - ▶ 4 GiB addressable

Register file (integer)

summary

Symbolic name	Number	Usage
zero	x0	Constant 0
ra	x1	Return address (used by function calls)
sp	x2	Stack pointer
gp	x3	Pointer to global area
tp	x4	Thread pointer
t0...t2	x5-x7	Temporary (NO preserved across calls)
s0/fp	x8	Temporary (preserved across calls) / Frame pointer
s1	x9	Temporary (preserved across calls)
a0...a1	x10...11	Input function arguments / return values
a2...a7	12...x17	Input function arguments
s2... s11	x18...x27	Temporary (preserved across calls)
t3...t6	x28...x31	Temporary (NO preserved across calls)

- ▶ There are 32 registers
 - ▶ Size: 4 bytes (1 word)
 - ▶ Dual name: logical and numerical (starts with **x** at the beginning)
- ▶ Use convention
 - ▶ Reserved
 - ▶ Arguments
 - ▶ Results
 - ▶ Temporary
 - ▶ Pointers

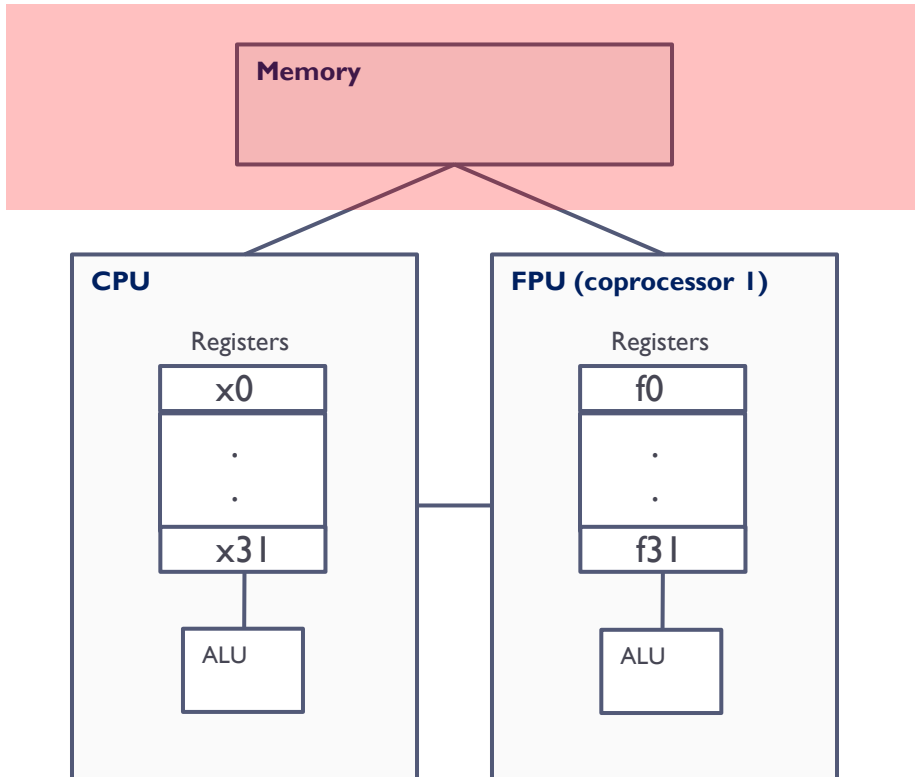
Register file (floating point)

summary

Logical name	Numeric name	Description
ft0-ft7	f0 ... f7	Temporals (like t...)
fs0-fs1	f8 ... f9	Temporals preserved (like s...)
fa0-fa1	f10 ... f11	Arguments/return (like a...)
fa2-fa7	f12 ... f17	Arguments (like a...)
fs2-fs11	f18 ... f27	Temporals preserved (like s...)
ft8-ft11	f28 ... f31	Temporals (like t...)

- ▶ There are 32 registers
- ▶ In R32F (simple precision) registers are 32-bits (4 bytes)
- ▶ In R32D (double precision) registers are 64-bits (8 bytes) and can store:
 - ▶ Single-precision values in the lower 32 bits of the register
 - ▶ Double precision values in all 64 bits of the register

RISC-V 32



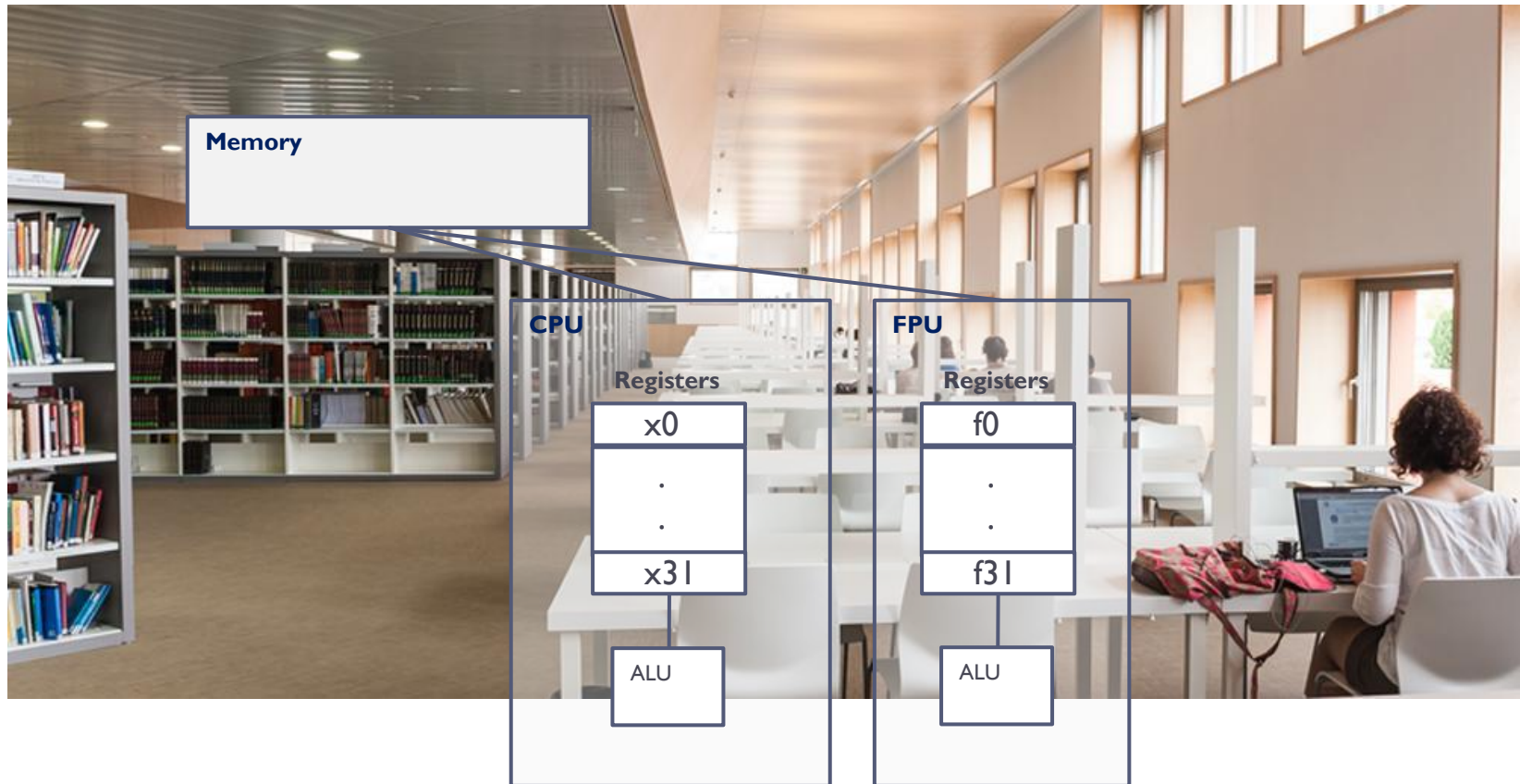
▶ Registers

- ▶ Inside the processor
- ▶ Instructions work with values in registers
- ▶ 32 registers of 32 bits

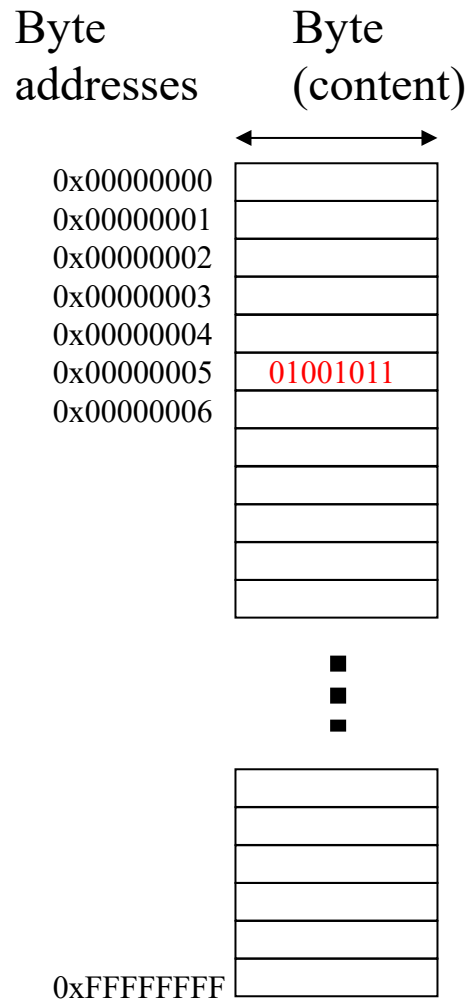
▶ Memory

- ▶ Outside the processor
- ▶ Data exchange to/from registers
- ▶ More capacity but longer access time than registers
- ▶ 32-bit addresses
 - ▶ 4 GiB addressable

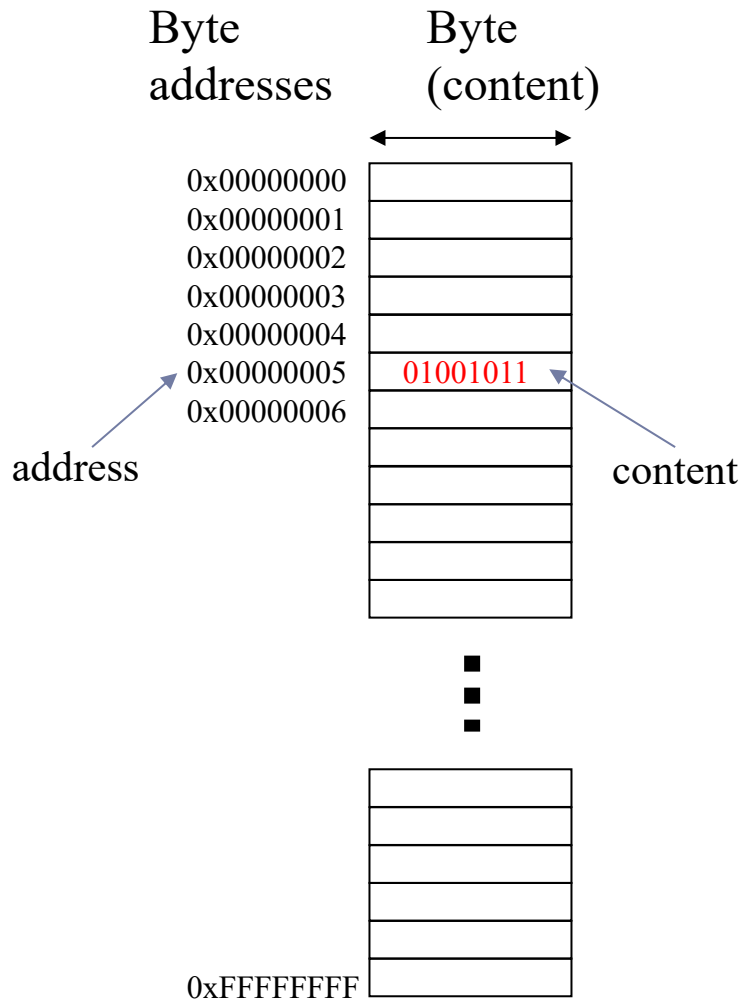
RISC-V 32 (memory and registers)



RISC-V 32 memory model



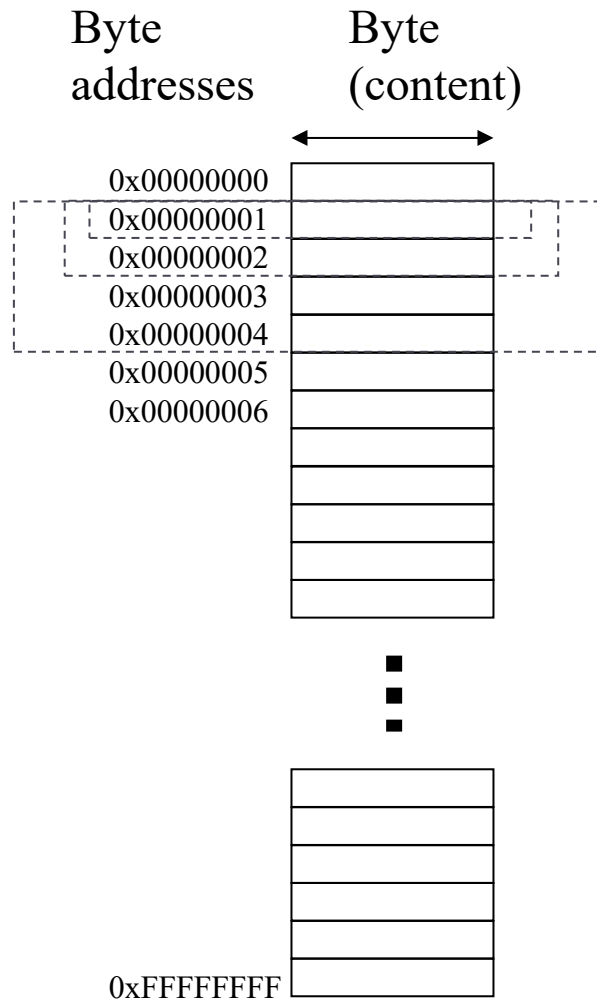
RISC-V 32 memory model



Memory is addressed at byte level:

- **32-bit addresses**
- **Content of each address: one byte**
- **Addressable space: 2^{32} bytes = 4 GB**

RISC-V 32 memory model



Memory is addressed at byte level:

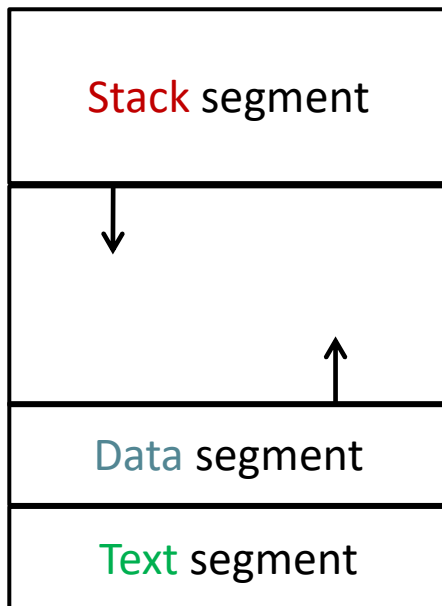
- **32-bit addresses**
- Content of **each address: one byte**
- Addressable space: 2^{32} bytes = 4 GB

Access can be to:

- Individual bytes
- Words (4 consecutive bytes)
- ~~Half-word (2 consecutive bytes)~~

Memory layout for a process

- ▶ The memory space is divided in logic segments in order to organize the content:

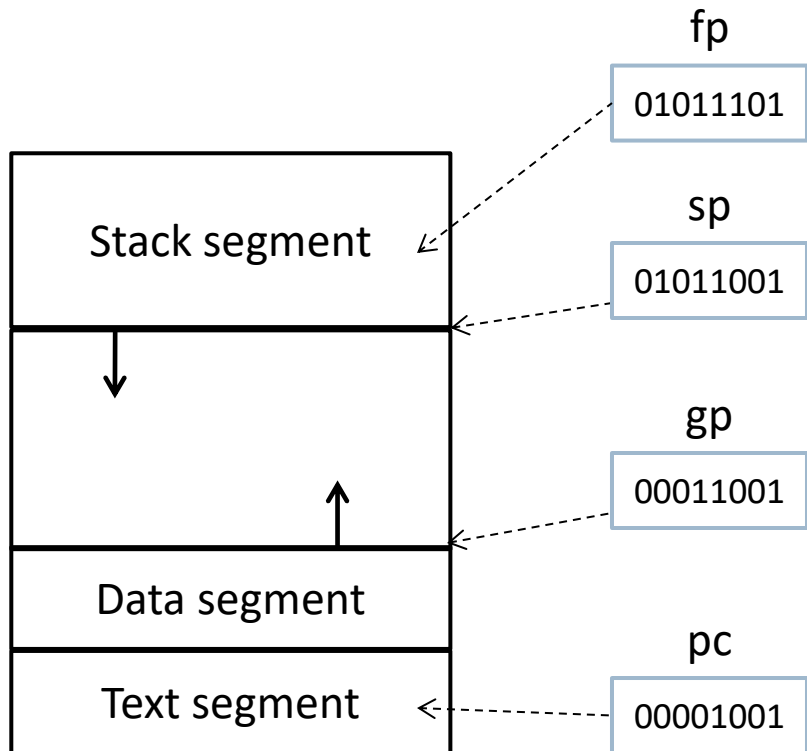


```
// global variables
int a;

main ()
{
    // local variables
    int b;

    // code (text)
    return a + b;
}
```

Memory layout for a process



- ▶ The memory space is divided in logic segments in order to organize the content:

- ▶ Stack segment
 - ▶ Local variables
 - ▶ Function contexts
- ▶ Data segments
 - ▶ Global variables
 - ▶ Static variables
- ▶ Code segment (text)
 - ▶ Program code

Example: hello world...

hello.s

.data

msg_hola: **.string** "hello !\n"

.text

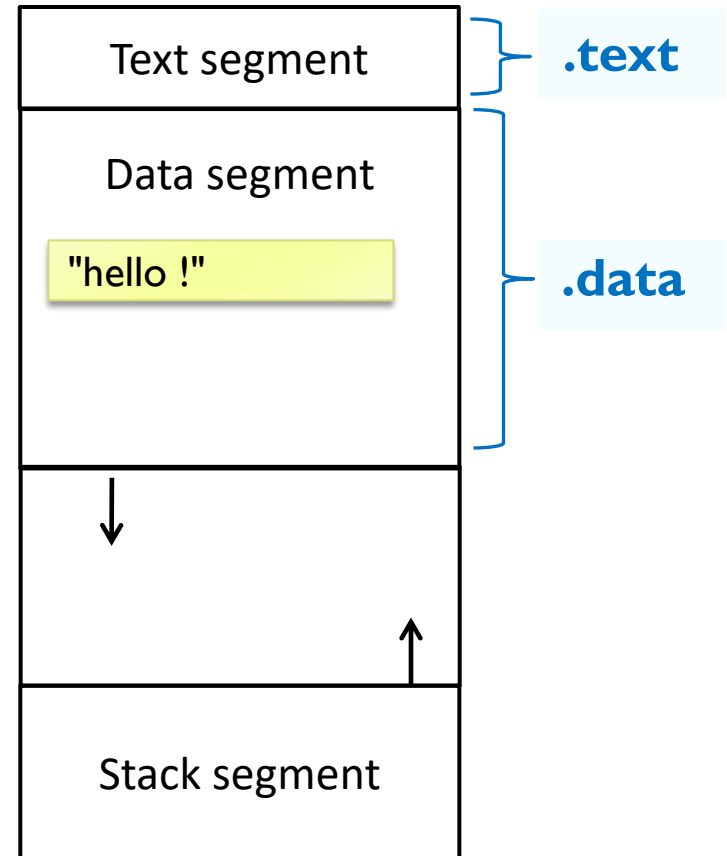
main:

printf("hello !\n") ;

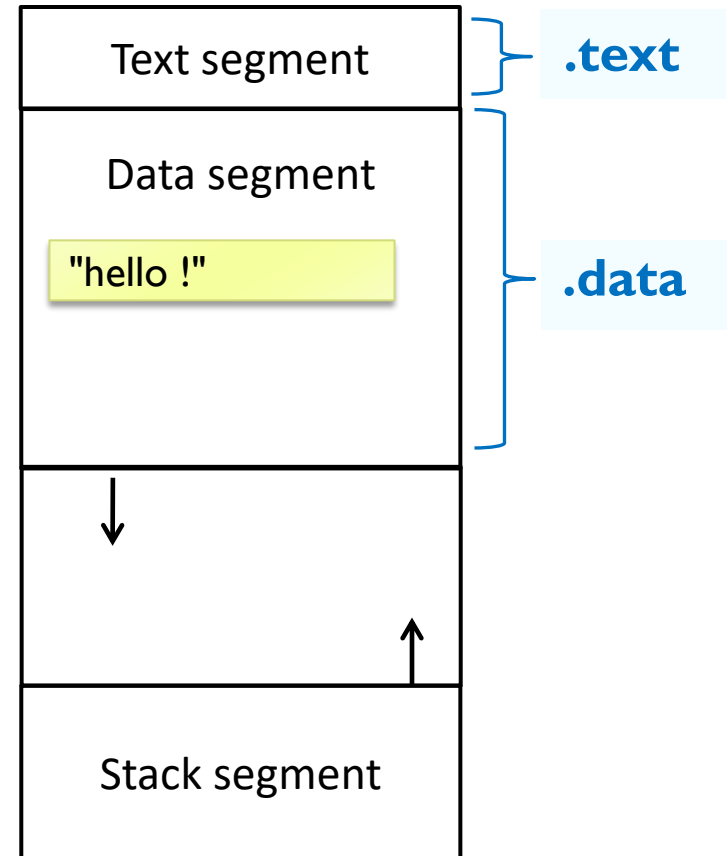
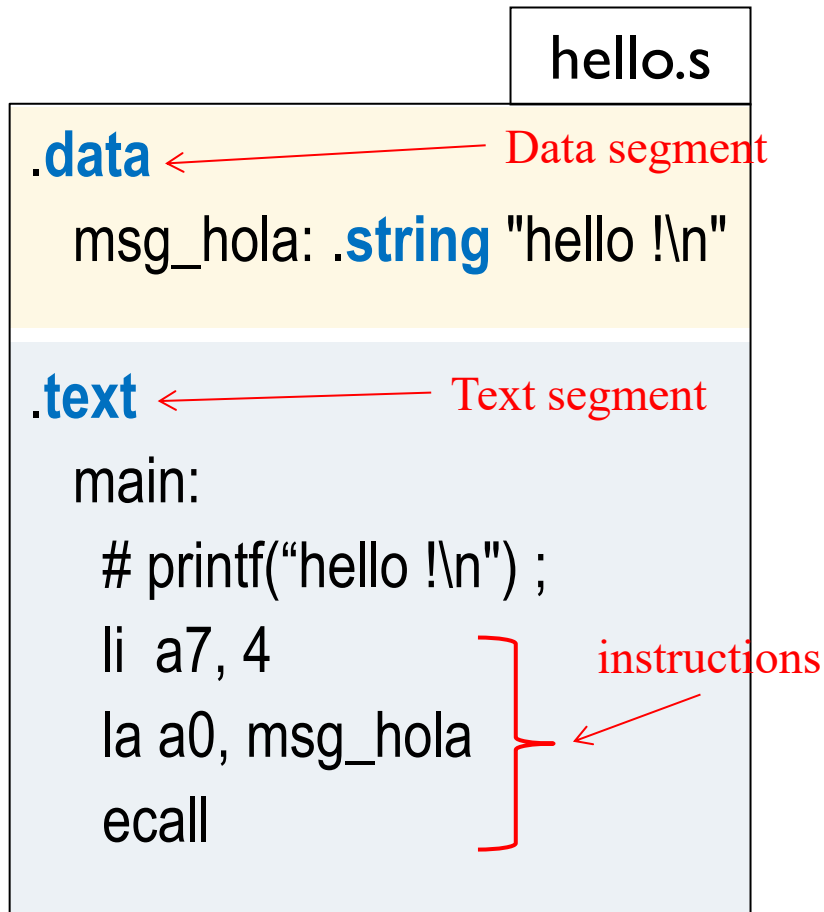
li a7, 4

la a0, msg_hola

ecall



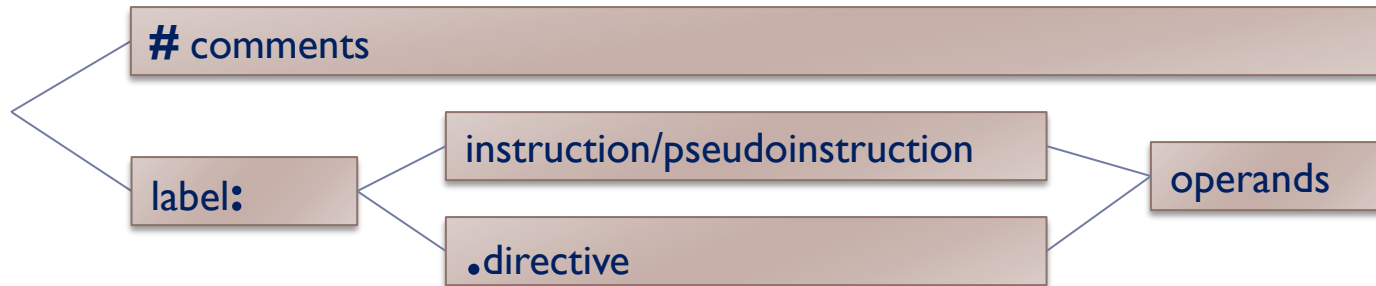
Example: hello world...



Contents

1. Basic concepts on assembly programming
2. RISC-V 32 assembly, memory model and data representation
 - ▶ RISC-V: registers and memory
 - ▶ **Assembler directives**
 - ▶ System services
 - ▶ Memory access instructions
3. Instruction formats and addressing modes
4. Floating point in assembly
5. Procedure calls and stack convention

Example: hello world...



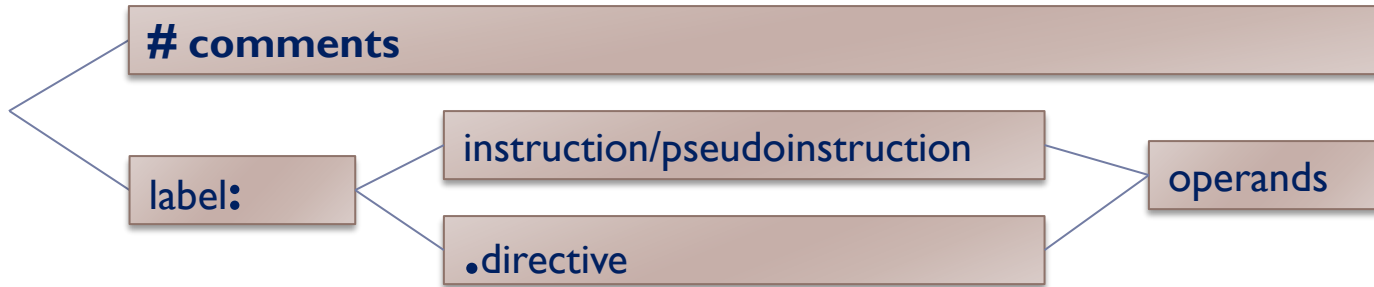
hello.s

```
.data
    msg_hola: .string "Hello world\n"

.text
main:
    # printf("Hello world\n") ;
    li a7, 4
    la a0, msg_hola
    ecall
```

Assembly:

Comments with



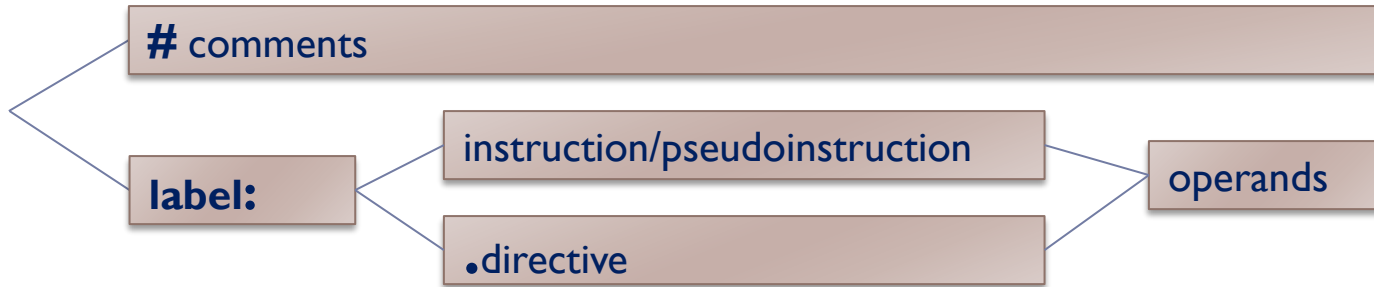
hello.s

```
.data
    msg_hola: .string "Hello world\n"

.text
main:
    # printf("Hello world\n") ;
    li a7, 4
    la a0, msg_hola
    ecall
```

Assembly:

labels ends with :



hello.s

```
.data
msg_hola: .string "Hello world\n"
```

```
.text
```

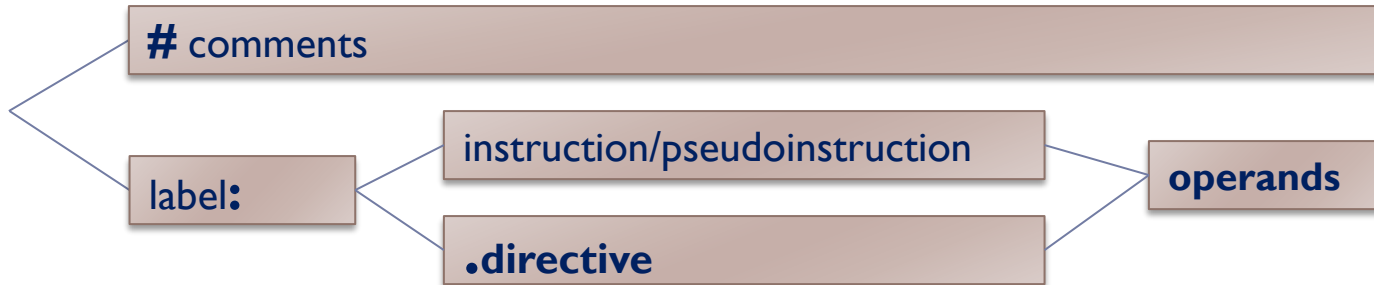
```
main:
# printf("Hello world\n") ;
li a7, 4
la a0, msg_hola
ecall
```

msg_hola: represents the memory address
where the string starts to be stored

main: represents the memory address where the
first main instruction is stored

Assembly:

Directives begins with .



hello.s

```
.data
    msg_hola: .string "Hello world\n"

.text
    main:
        # printf("Hello world\n") ;
        li a7, 4
        la a0, msg_hola
        ecall
```

Assembly: directives

Directives	Description
.data	Next elements will go to the data segment
.text	Next elements will go to the code segment
.ascii <i>"string"</i>	String definition <u>without</u> '\0' ending terminator ('\0' = 0)
.string <i>"string"</i>	String definition <u>with</u> '\0' ending terminator ('\0' = 0)
.byte 1, 2, 3	Bytes stored in memory consecutively
.half 300, 301, 302	Half-words stored in memory consecutively
.word 800000, 800001	Words stored in memory consecutively
.float 1.23, 2.13	Floats stored in memory consecutively
.double 3.0e21	Doubles stored in memory consecutively
.zero 10	Allocates a space of 10 bytes in the current segment
.align <i>n</i>	Align next element to a address multiple of 2^n

Definition of static data (at compile time)

label (address)

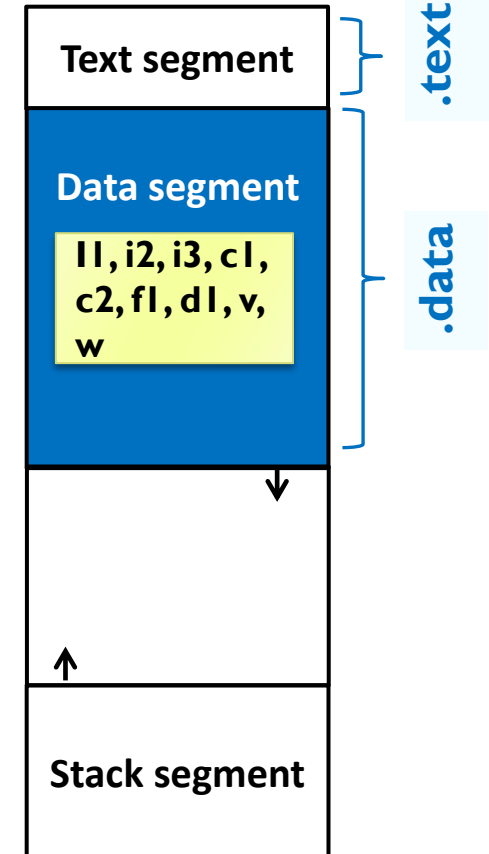
Data type (directive)

value

```
.data
cadena .string "Hello world\n"
i1: .word 10      # int    i1=10
i2: .word -5      # int    i2=-5
i3: .half 300     # short  i3=300
c1: .byte 100     # char   c1=100
c2: .byte 'a'     # char   c2='a'
f1: .float 1.3e-4 # float  f1=1.3e-4
d1: .double .001  # double d1=0.001

# int v[3]={0,-1,0xffffffff}; int w[100];
v: .word 0, -1, 0xffffffff
w: .zero 400
```

0x00



Representation of basic data types (1/3)

```
// boolean
bool_t b1 ;
bool_t b2 = false ;

// character
char c1 ;
char c2 = 'x' ;

// integers
int  res1 ;
int  op1 = -10 ;

// floating point
float  f0 ;
float  f1 = 1.2 ;
double d2 = 3.0e10 ;
```

```
.data

# boolean
b1:    .zero    1           # 1 bytes
b2:    .byte    0           # 1 byte

# character
c1:    .byte           # 1 bytes
c2:    .byte 'x'       # 1 bytes

# integers
res1:  .zero    4           # 4 bytes
op1:   .word    -10         # 4 bytes

# floating point
f0:    .float           # 4 bytes
f1:    .float    1.2       # 4 bytes
d2:    .double   3.0e10    # 8 bytes
```

Representation of basic data types (2/3)

```
// strings
char c1[10] ;
char ac1[] = "hola" ;
char ac2[] = ['h','o','l','a'] ;
```

.data

```
# strings
c1:   .zero    10      # 10 byte
ac1:  .string  "hola"  # 5 bytes (!)
ac1:  .ascii   "hola"  # 4 bytes
ac2:  .byte    'h', 'o', 'l', 'a'
```

ac1:	...	
	'h'	0x0108
	'o'	0x0109
	'l'	0x010a
	'a'	0x010b
	0	0x010c
	...	0x010d

ac2:	...	
	'h'	0x0108
	'o'	0x0109
	'l'	0x010a
	'a'	0x010b
	...	0x010c
	...	0x010d

Representation of basic data types (3/3)

```
// vectors
int vec[5] ;
int mat[2][3] = {{11,12,13},
                 {21,22,23}};
```

.data

```
# vectors
vec:  .zero 20      # 5 elem.*4 bytes
mat:  .word 11, 12, 13
      .word 21, 22, 23
```

mat:

	...	
11		
12		
13		
21		
22		
23		
...		

0x0108

0x010c

0x0110

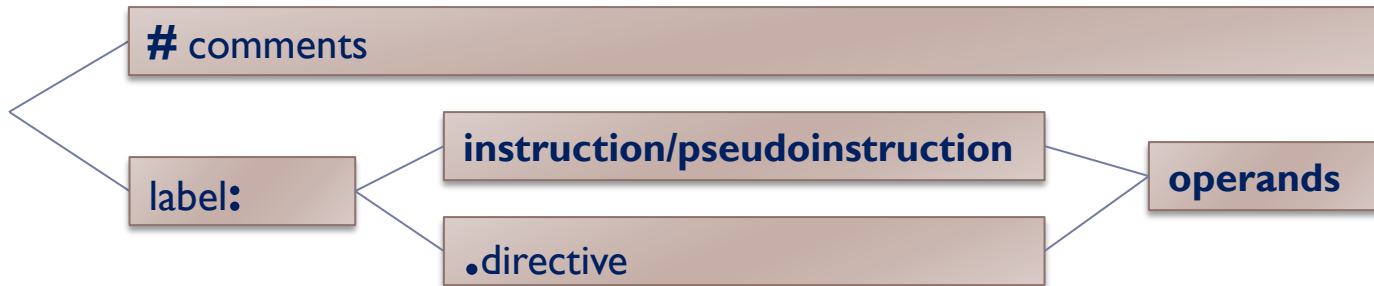
0x0114

0x0118

0x011c

Assembly:

Instructions vs pseudoinstructions



hello.s

```
.data
    msg_hola: .string "Hello world\n"

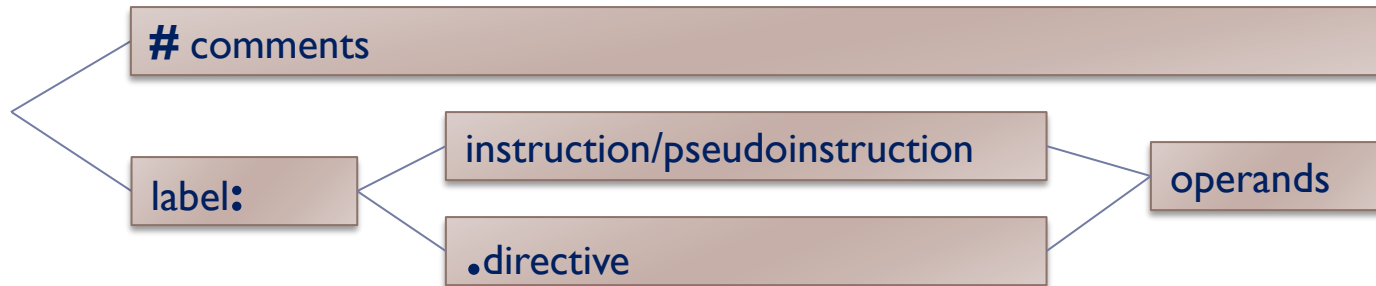
.text
main:
    # printf("Hello world\n") ;
    li a7, 4
    la a0, msg_hola
    ecall
```

Instructions and pseudoinstructions

- ▶ An assembly **instruction** corresponds to a single machine instruction:
 - ▶ Occupies 32 bits in RISC-V 32
 - ▶ E.g.: `addi t1, t0, 4`
- ▶ A pseudo-instruction can be used in an assembler program but does not correspond to any machine instruction:
 - ▶ In the assembly process, they are replaced by the sequence of machine instructions that perform the same functionality.

E.g.: `mv t1, t2` is replaced by: `add t1, x0, t2`
`li t1, 0x00800010` is replaced by: `lui t1, 0x00800` (20 bits)
`addi t1, t1, 0x010` (12 bits)

Assembly: summary



hello.s

```
.data
    msg_hola: .string "Hello world\n"

.text
main:
    # printf("Hello world\n") ;
    li a7, 4
    la a0, msg_hola
    ecall
```

Contents

1. Basic concepts on assembly programming
2. RISC-V 32 assembly, memory model and data representation
 - ▶ RISC-V: registers and memory
 - ▶ Assembler directives
 - ▶ System services
 - ▶ Memory access instructions
3. Instruction formats and addressing modes
4. Floating point in assembly
5. Procedure calls and stack convention

System calls

- ▶ Many simulators include a small "operating system"
 - ▶ CREATOR provides 12 services.
- ▶ How to invoke:
 - ▶ Call code in register **a7**
 - ▶ Other arguments on specific registers (eg.: **a0**, **fa0**)
 - ▶ Invocation by the **ecall** instruction

```
# printf("hello world\n")  
li  a7, 4  
la  a0, msg_hola  
ecall
```

System calls

Service	Call code (a7)	Arguments (a0, a1 and fa0)	Results (a0 and fa0)
print_int	1	a0 = integer	
print_float	2	fa0 = float	
print_double	3	fa0 = double	
print_string	4	a0 = string	
read_int	5		integer in a0
read_float	6		float in fa0
read_double	7		double in fa0
read_string	8	a0 = buffer, a1 = length	
sbrk	9	a0 = amount	address in a0
exit	10		
print_char	11	a0 (ASCII code)	
read_char	12		a0 (ASCII code)

Example: print “hello world”...

hello.s

.data

msg_hola: .string "hello world\n"

.text

main:

printf("hello world\n") ;

li a7, 4

la a0, msg_hola

ecall

Service	Call code (a7)	Arguments (a0, a1 and fa0)
print_int	1	a0 = integer
print_float	2	fa0 = float
print_double	3	fa0 = double
print_string	4	a0 = string

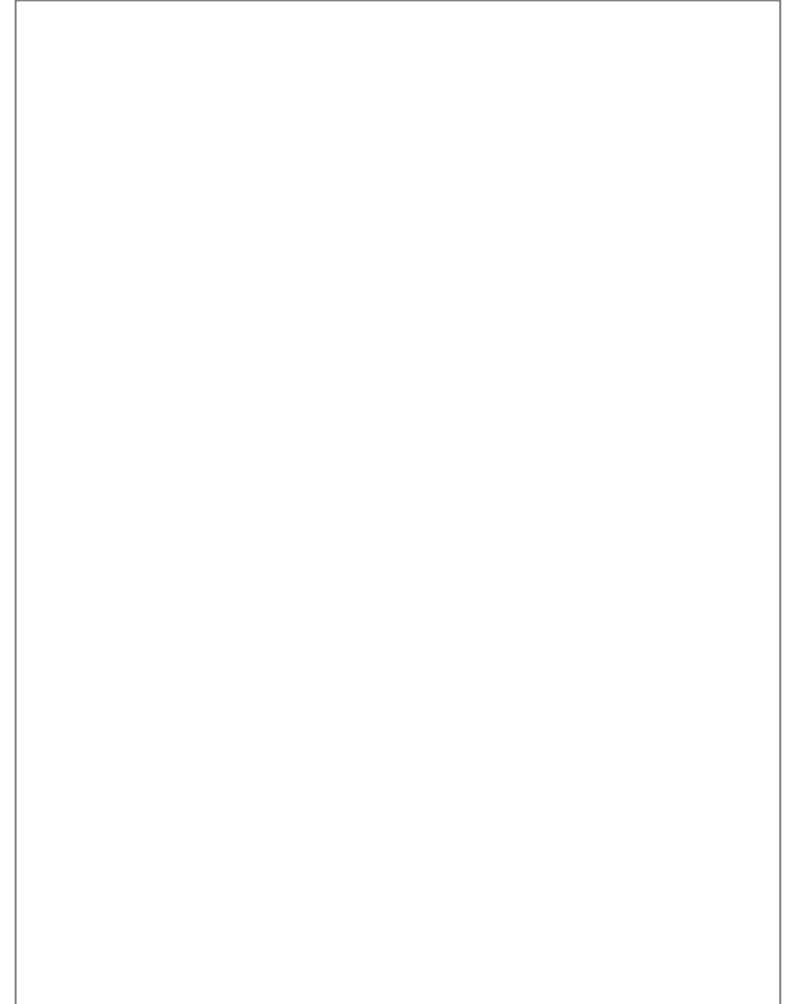
Operating system
invocation
instruction

Exercise: read integer, print integer

. . .

```
readInt(&value) ;  
value = value + 1 ;  
printInt(value) ;
```

. . .



Exercise: read integer, print integer

solution

CREATOR

WepSIM

. . .

```
readInt(&value) ;  
value = value + 1 ;  
printInt(value) ;
```

. . .

Service	Call code (a7)	Arguments (a0, a1 and fa0)	Results (a0 and fa0)
print_int	1	a0 = integer	
print_float	2	fa0 = float	
print_double	3	fa0 = double	
print_string	4	a0 = string	
read_int	5		integer in a0

. . .

```
# readInt(&value)  
li a7, 5  
ecall  
mv t0, a0 # value in t0
```

```
# value = value + 1  
addi t0, t0, 1
```

```
# printInt  
mv a0, t0  
li a7, 1  
ecall
```

. . .

sbrk: dynamic data allocation (at runtime)

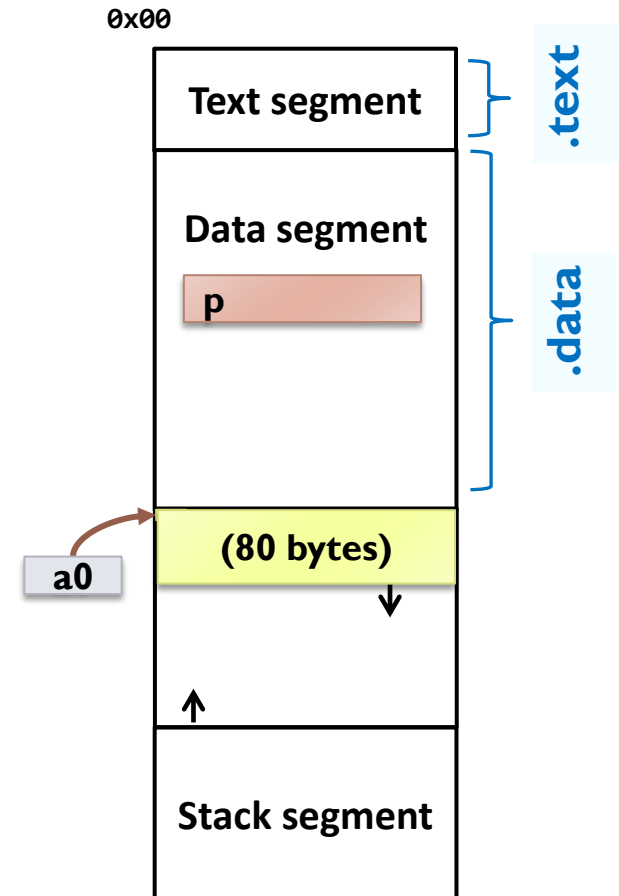
```
.data
# char *p; // pointer to a byte
p: .word 0x0

.text

# p = malloc(80); // 80 bytes
li a0, 80
li a7, 9 # system call
ecall

# the address is in a0
...

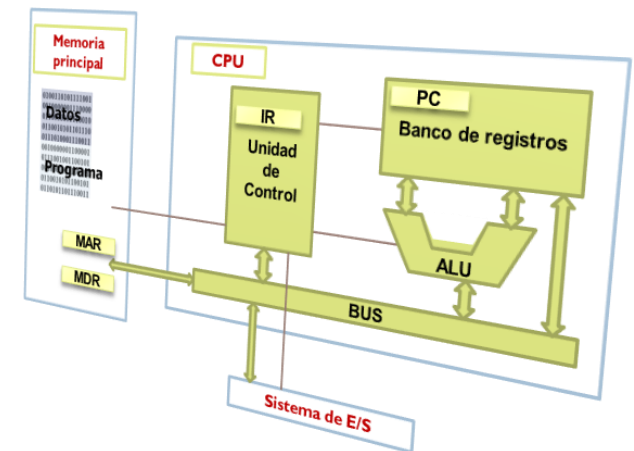
# free(p); // free 80 bytes
li a0, -80
li a7, 9 # system call
ecall
```



Contents

1. Basic concepts on assembly programming
2. RISC-V 32 assembly, memory model and data representation
 - ▶ RISC-V: registers and memory
 - ▶ Assembler directives
 - ▶ System services
 - ▶ Memory access instructions
3. Instruction formats and addressing modes
4. Floating point in assembly
5. Procedure calls and stack convention

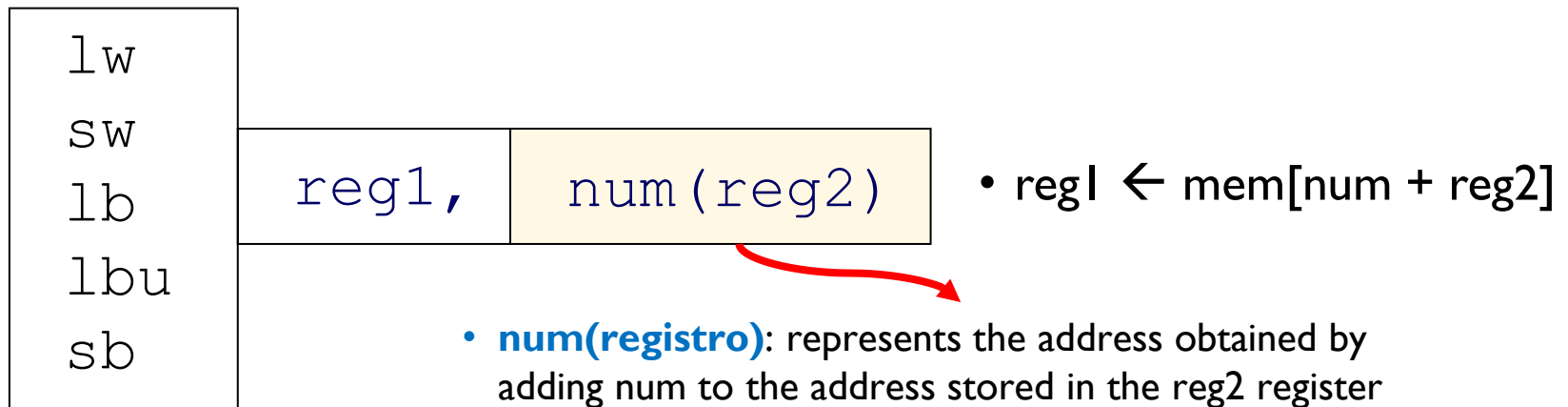
Access to memory



- ▶ Loading an address into a register:

```
la reg2, dir32 • reg2 ← dir32
```

- ▶ Memory access instructions (integers):

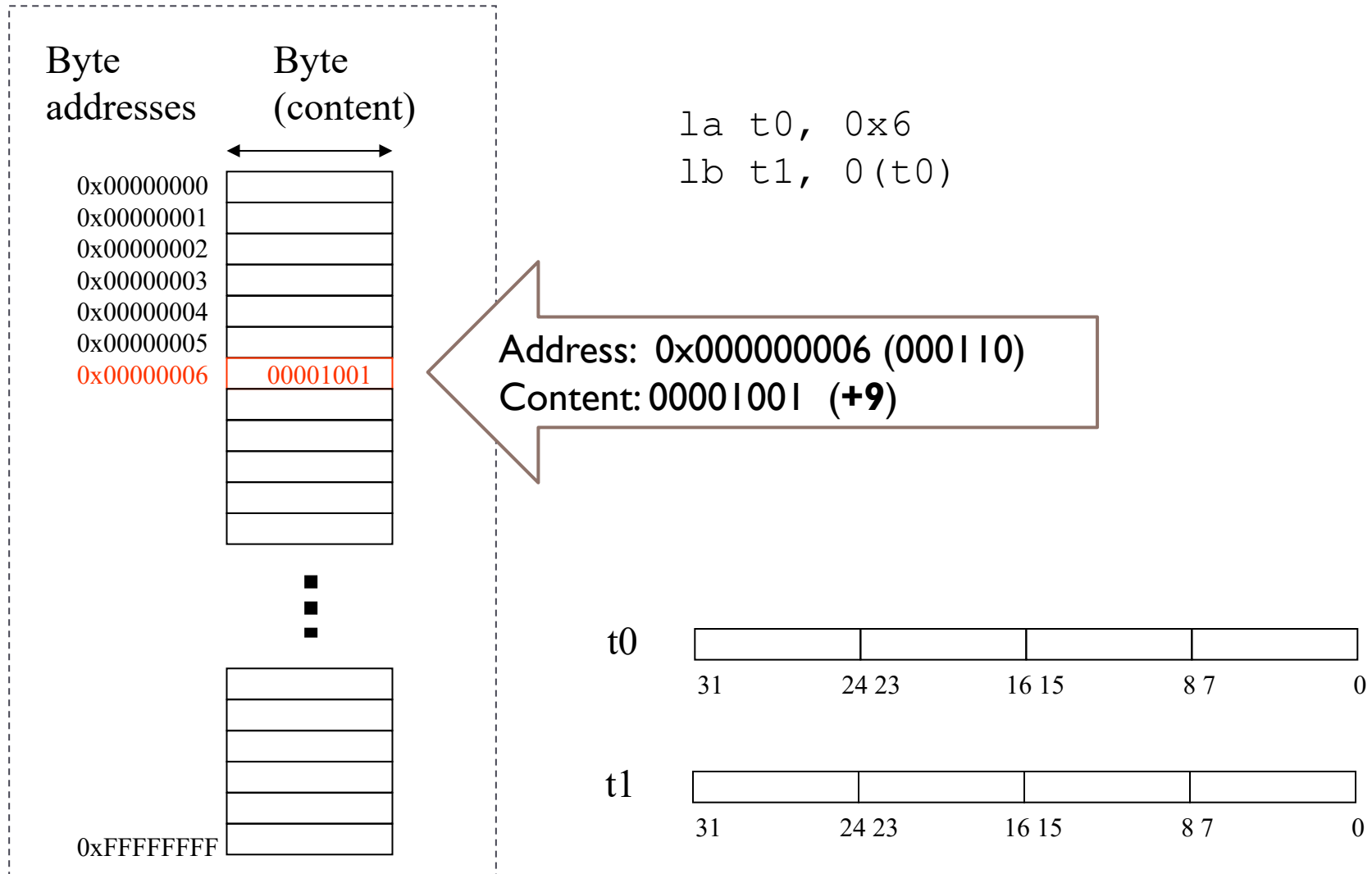


- **num(registro)**: represents the address obtained by adding num to the address stored in the reg2 register
 - num can be positive or negative
 - num format can be decimal or hexadecimal

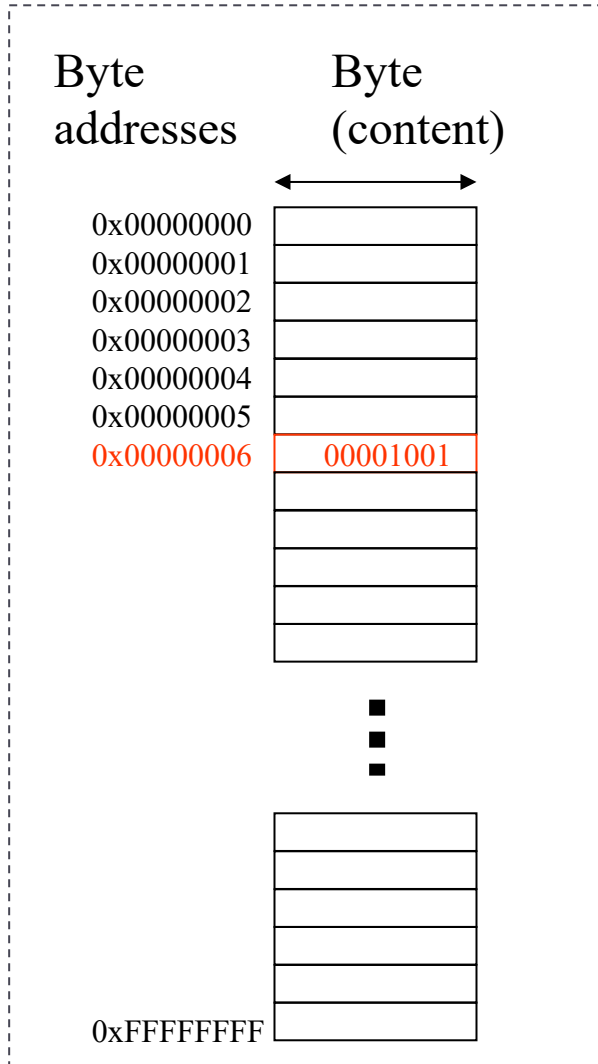
Memory operations

Memory access		
<code>la reg2, dir32</code>	<code>reg2 = dir32</code>	Load the 32-bit memory address dir32 into the reg2 register (pseudoinstruction)
<code>lw r1, num(r2)</code>	<code>r1 = Memory[r2+num]</code>	Accesses a stored word from memory location r2+num and saves it to r1
<code>sw r1, num(r2)</code>	<code>Memory[r2+num] = r1</code>	Stores from memory location r2+num the word contained in r1
<code>lb r1, num(r2)</code>	<code>r1 = Memory [r2+num]</code>	Accesses a byte stored in memory location r2+num and saves it in the least significant byte of r1
<code>lbu r1, num(r2)</code>	<code>r1 = Memory [r2+num]</code>	Accesses a byte stored in memory location r2+num and saves it in the least significant byte of r1
<code>sb r1, num(r2)</code>	<code>Memory[r2+num] = r1</code>	Saves from memory location r2+num the least significant byte of r1

Access to bytes with lb (*load byte*)

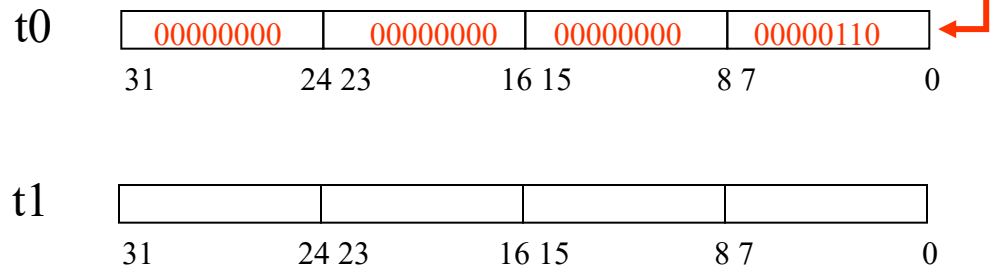


Access to bytes with lb (*load byte*)

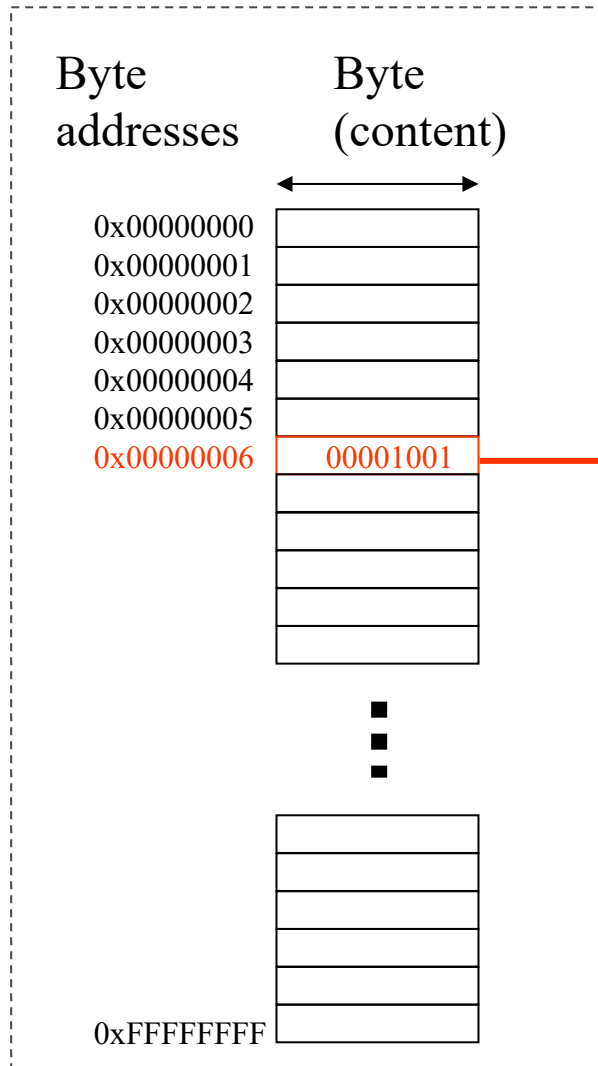


```
1a t0, 0x6
1b t1, 0(t0)
```

The address is copied, not the content



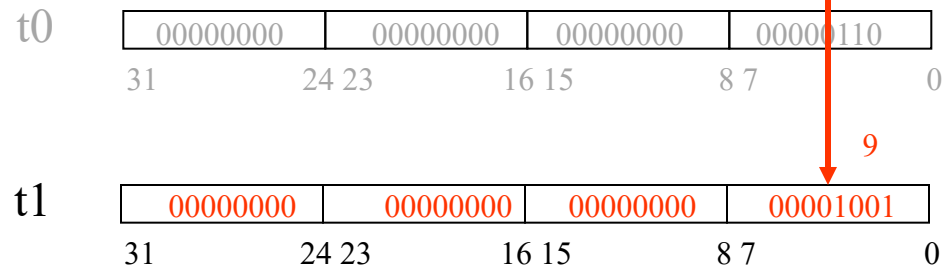
Access to bytes with lb (*load byte*)



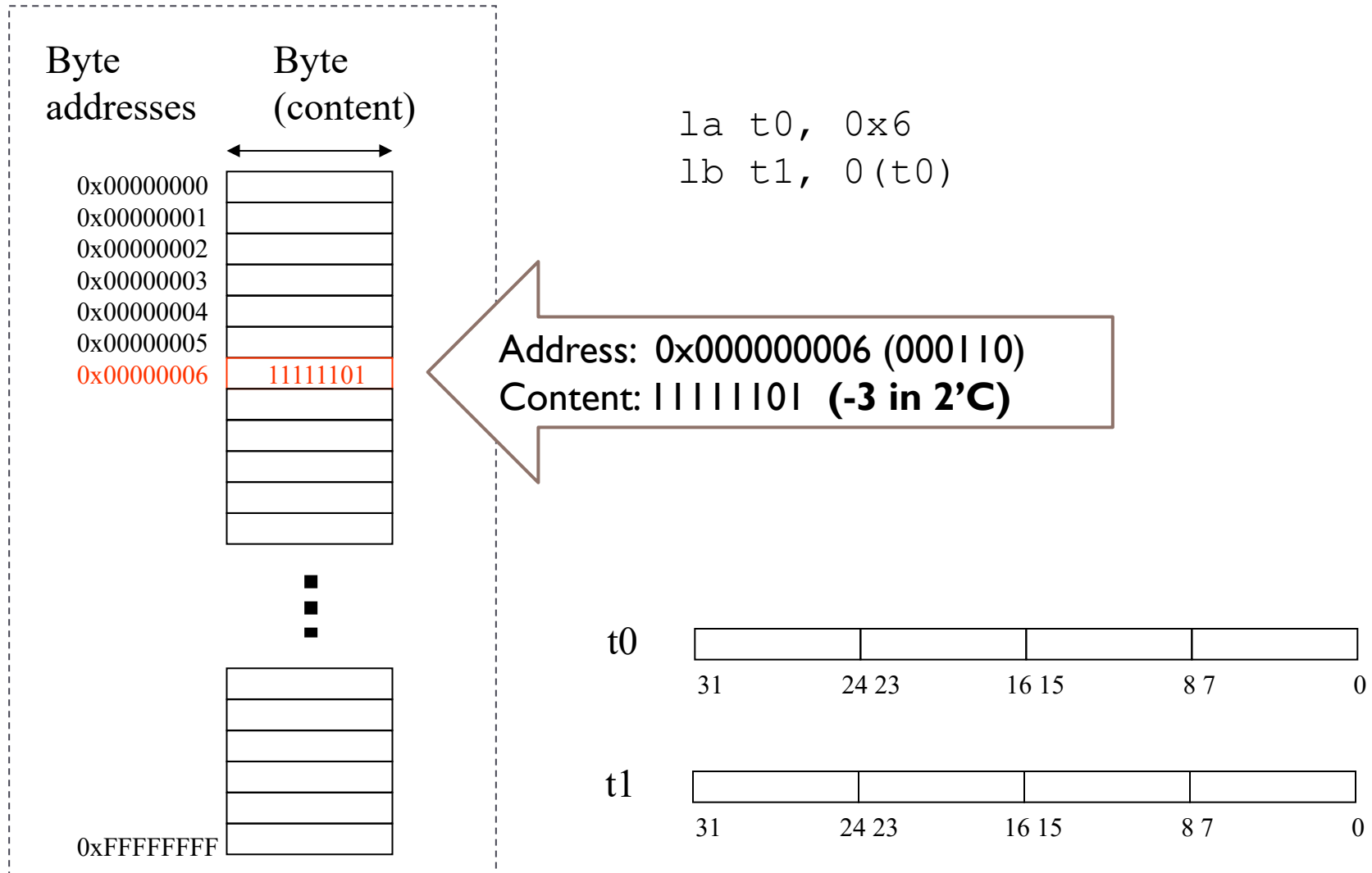
```
la t0, 0x6
```

```
lb t1, 0(t0)
```

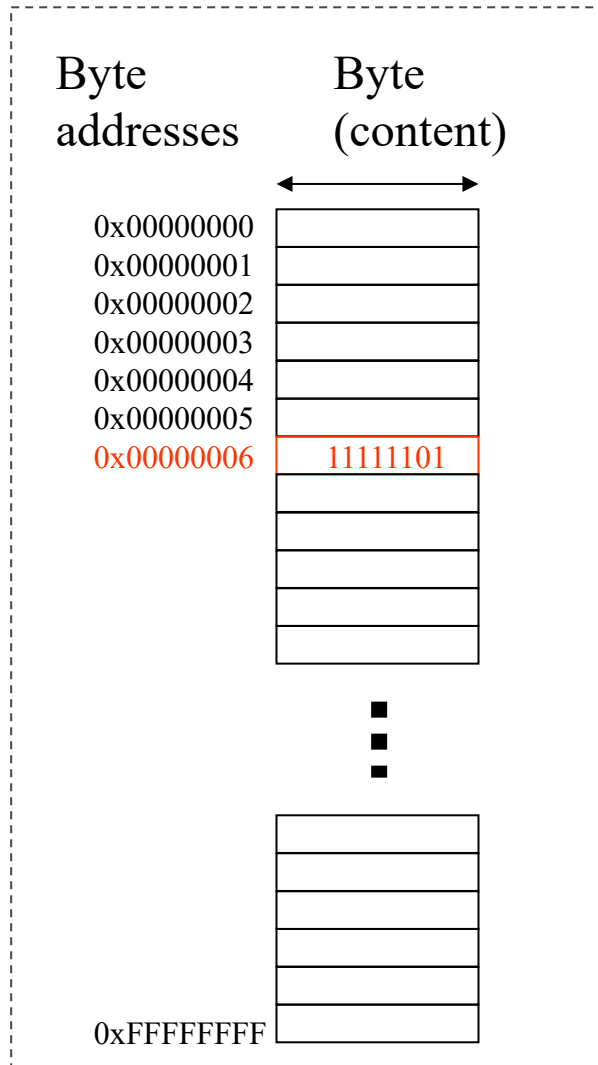
Copy the contents of 0x6 (9)



Access to bytes with lb (*load byte*)

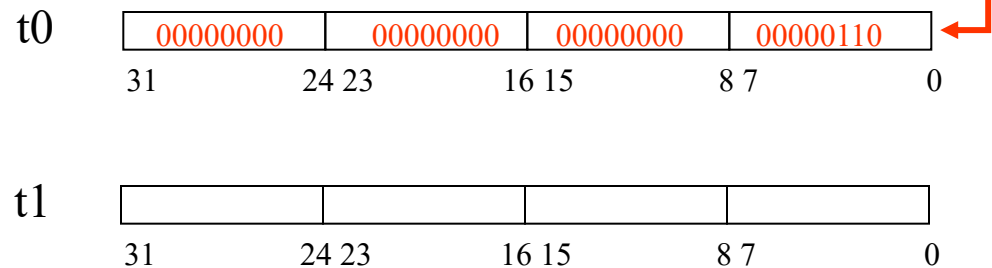


Access to bytes with lb (*load byte*)

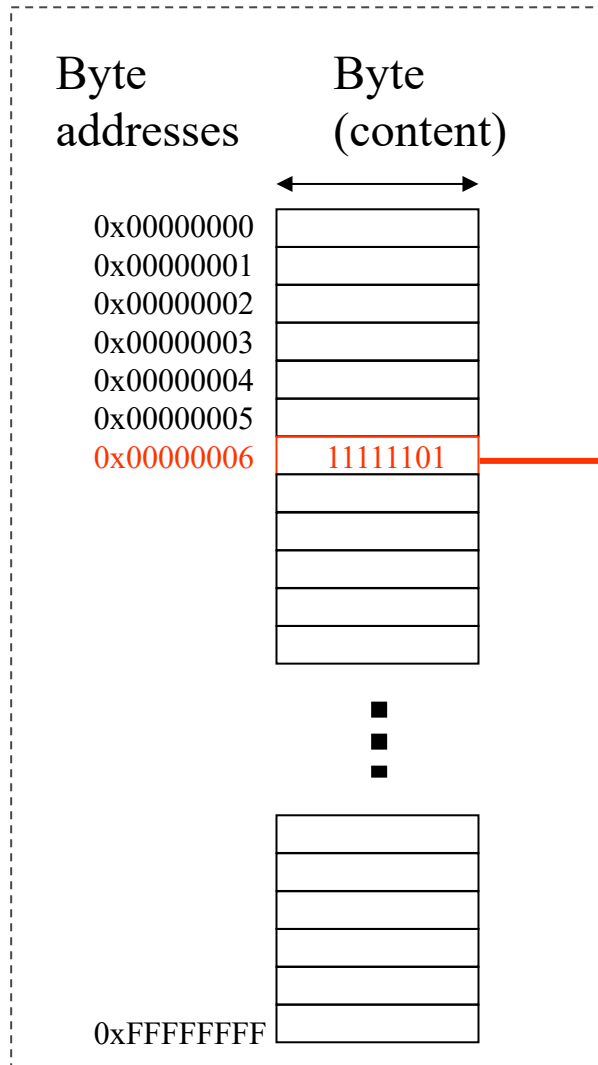


```
la t0, 0x6  
lb t1, 0(t0)
```

Copy 0x6 to t0



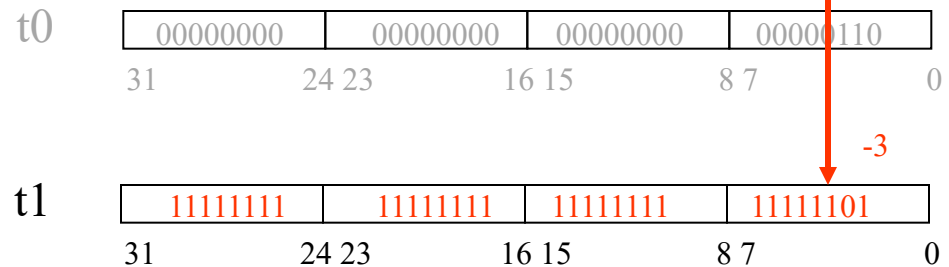
Access to bytes with lb (*load byte*)



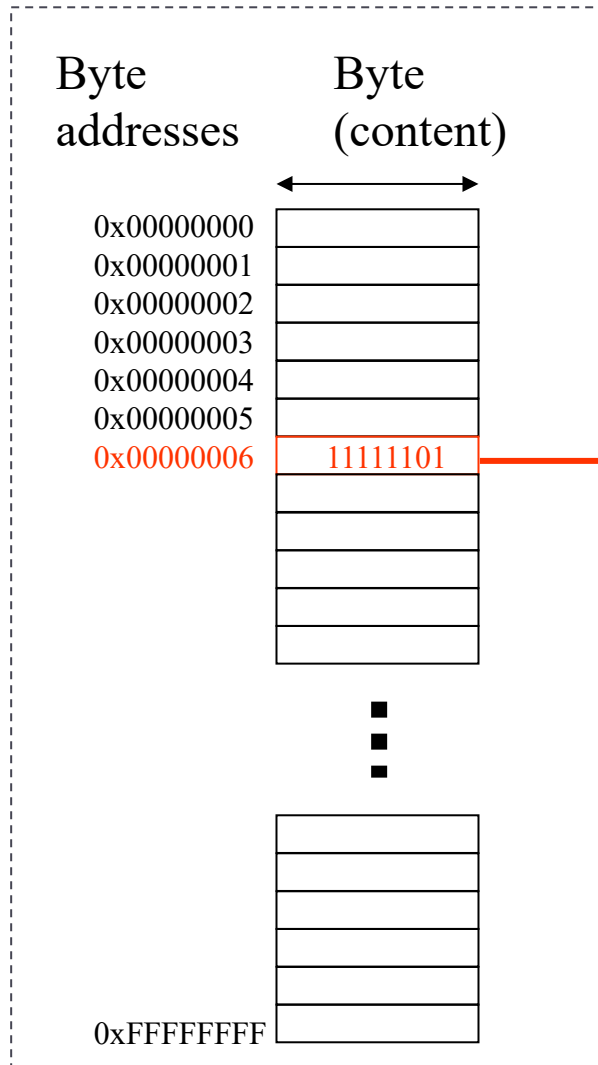
```
la t0, 0x6
```

```
lb t1, 0(t0)
```

Copy the contents of 0x6 (-3)



Access to bytes with lb (*load byte*)

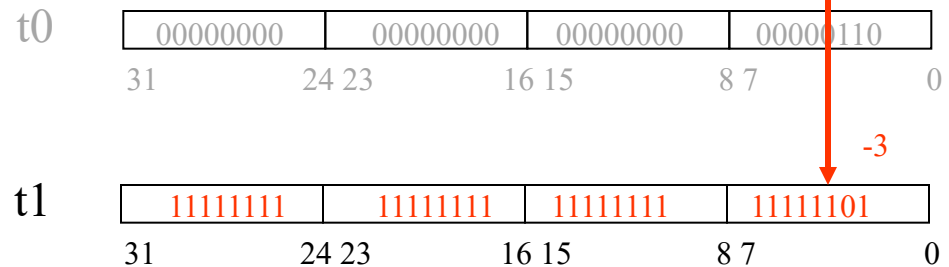


```
la t0, 0x6
```

```
lb t1, 0(t0)
```

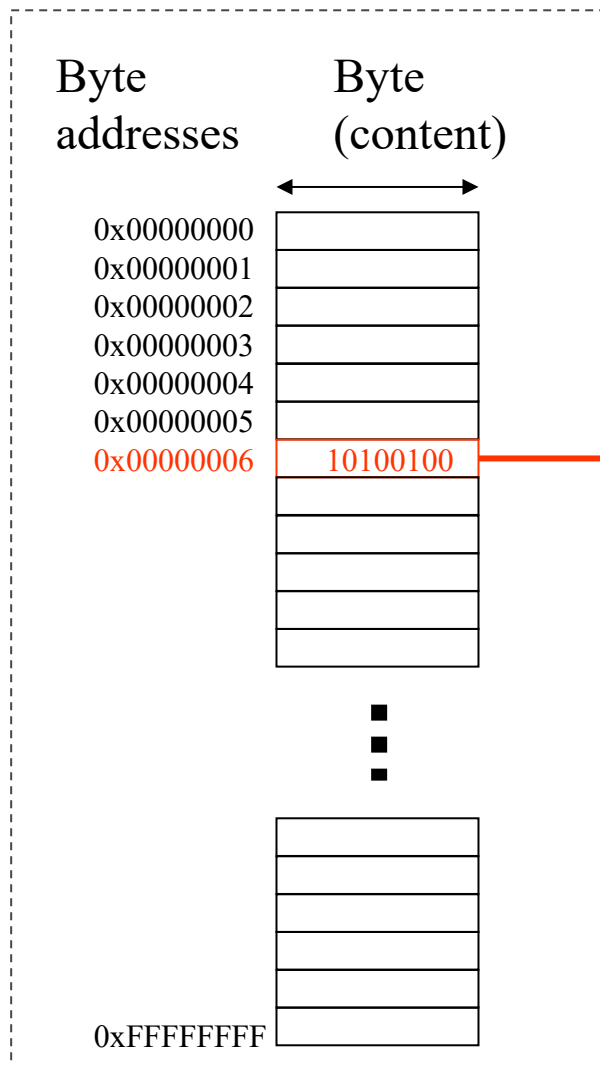
Copy the contents of 0x6 (-3)

The lb instruction keeps the sign:
does sign extension



Access to bytes with lb (*load byte*)

problems accessing characters

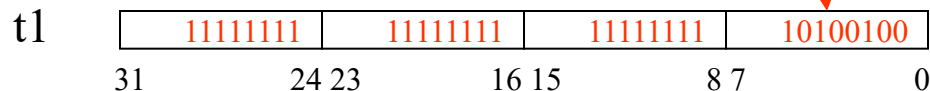


```
la t1, 0x6
lb t1, 0(t1)
```

Address: 0x000000006 (000110)

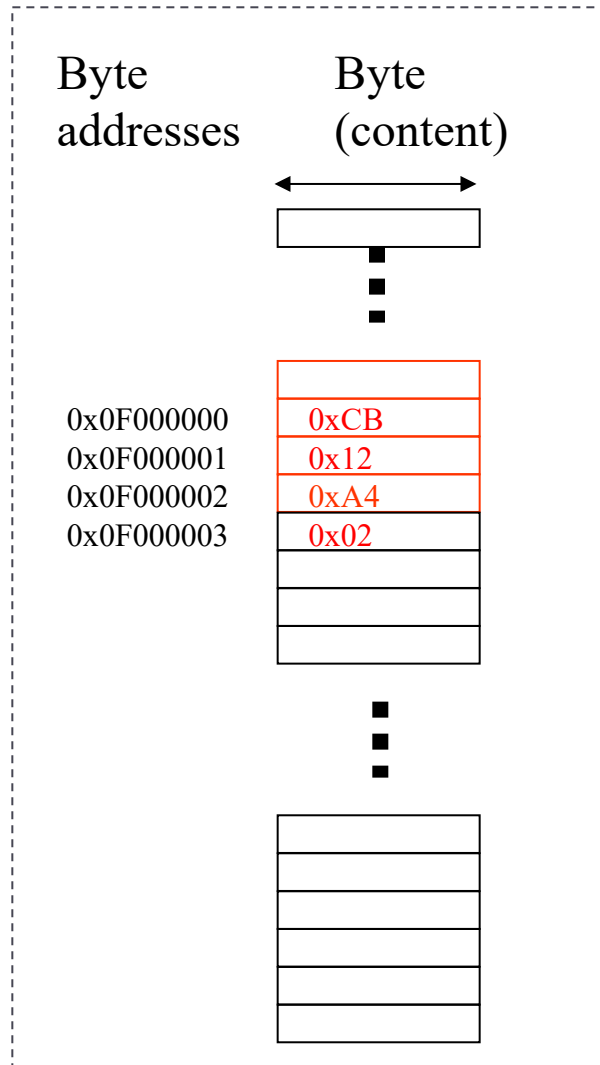
Content: 10100100 (ASCII code for ñ: 164)

```
la t1, 0x6
lb t1, 0(t1)
```

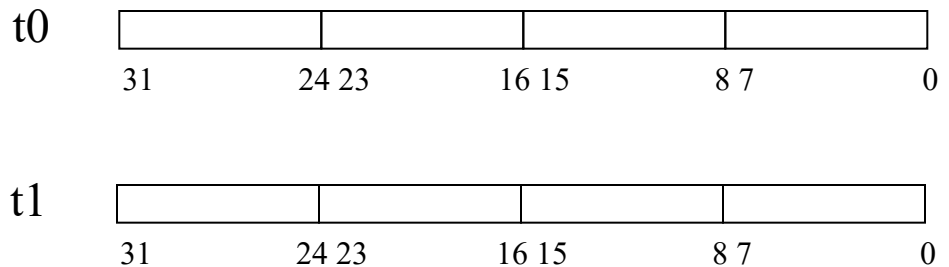


If lb is used (the sign is kept),
the content of t1 **does not** coincide with
the value 164 (ñ)

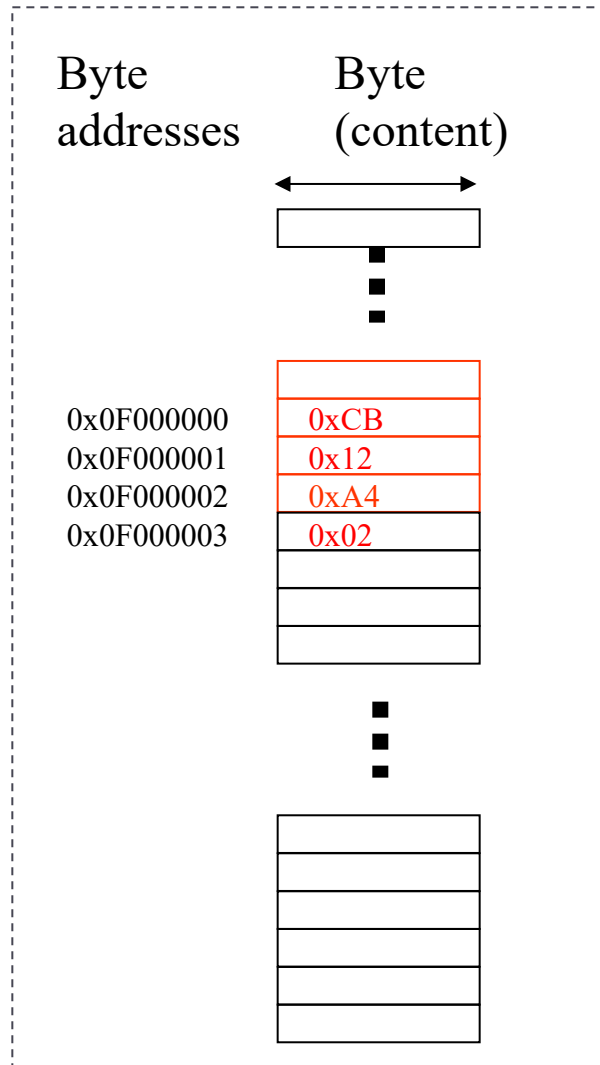
Access to bytes with lbu (*load byte unsigned*)



```
la    t0, 0x0F000002
lbu   t1, 0(t0)
```

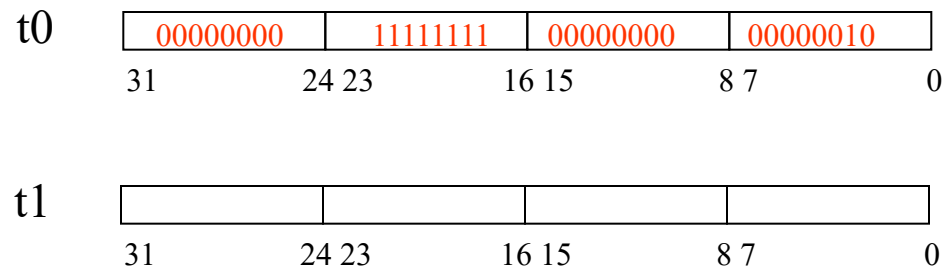


Access to bytes with lbu (*load byte unsigned*)

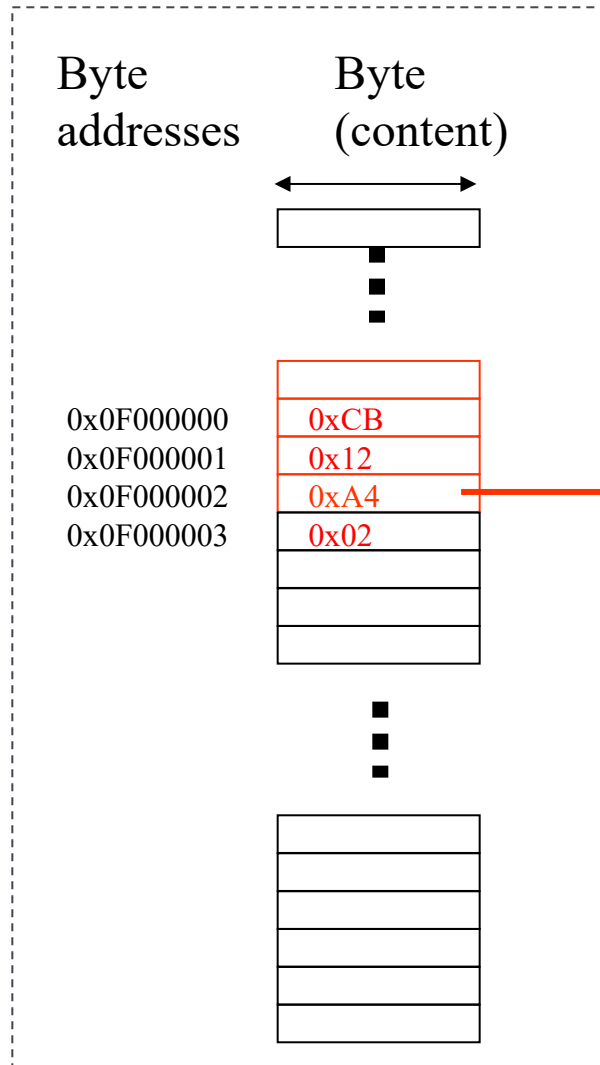


```
la    t0, 0x0F000002
lbu   t1, 0(t0)
```

The address is copied, not the content

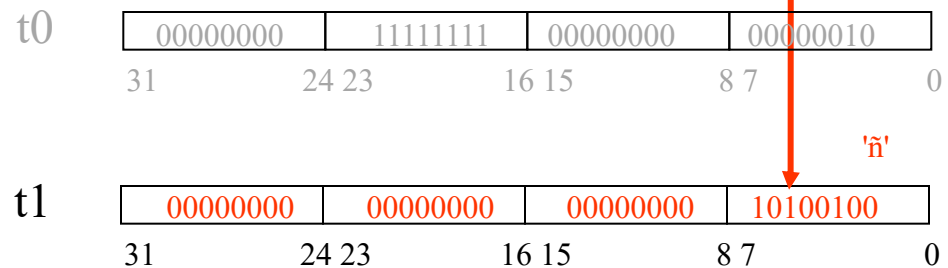


Access to bytes with lbu (*load byte unsigned*)

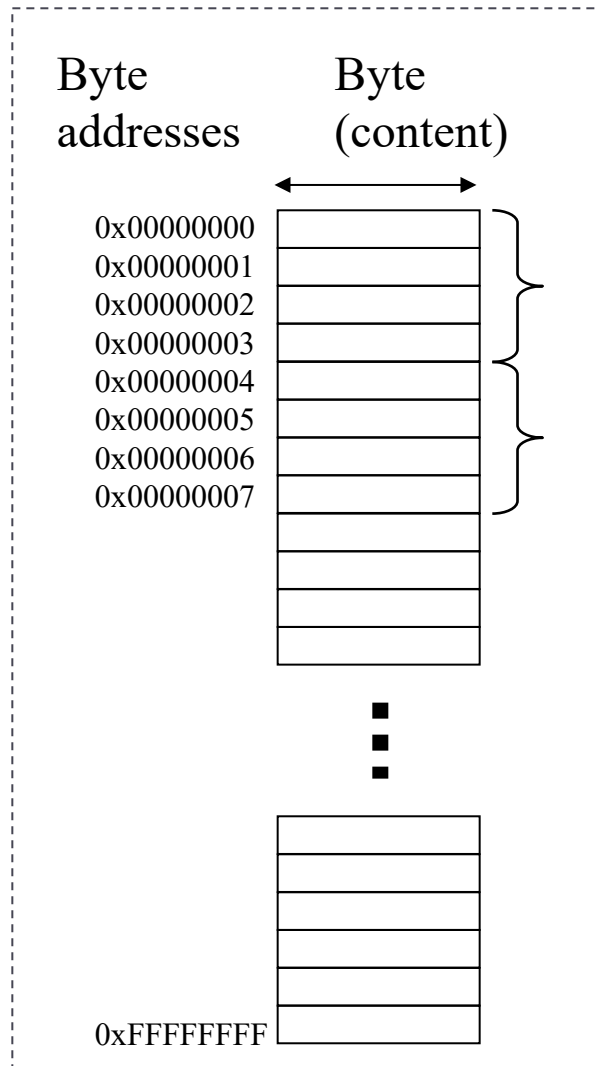


```
la    t0, 0x0F000002
lbu   t1, 0(t0)
```

The **byte** stored at position `0x0F000002` is copied (**without sign extension**)



Access to words



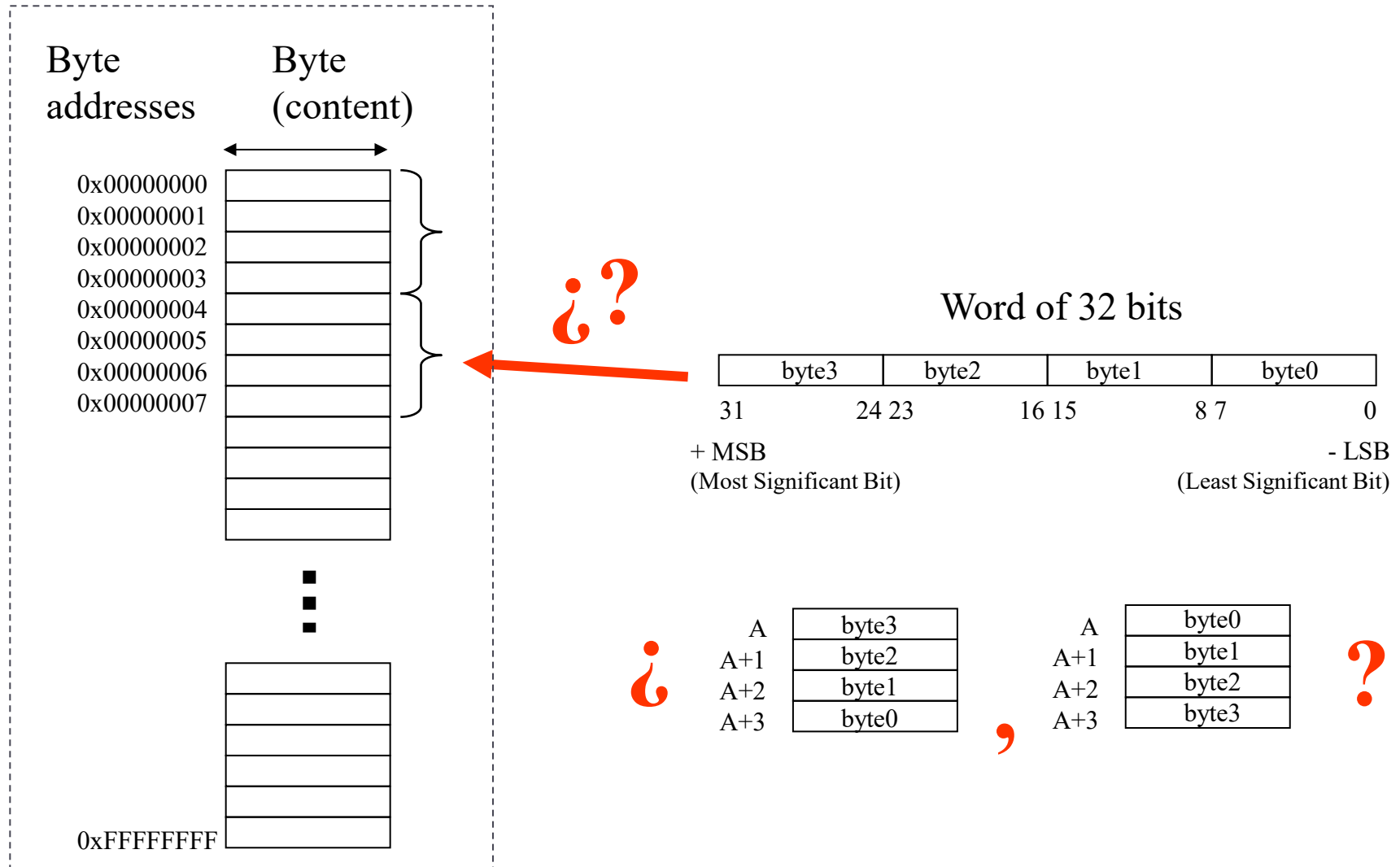
4 bytes make a word

Word stored from byte 0

Word stored from byte 4

The words (32 bits, 4 bytes) are stored using four consecutive memory locations, with the first location starting at an address multiple of 4

Access to words

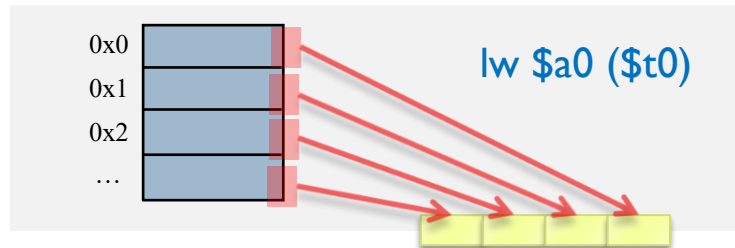


Storage of words in memory

byte order

► There are 2 types of byte order:

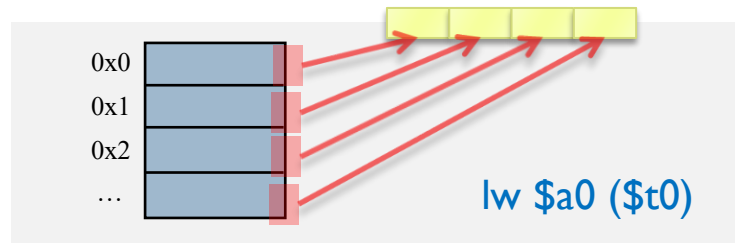
► Little-endian (‘small’ address ends the word...)



AMD



► Big-endian (‘big’ address ends the word...)



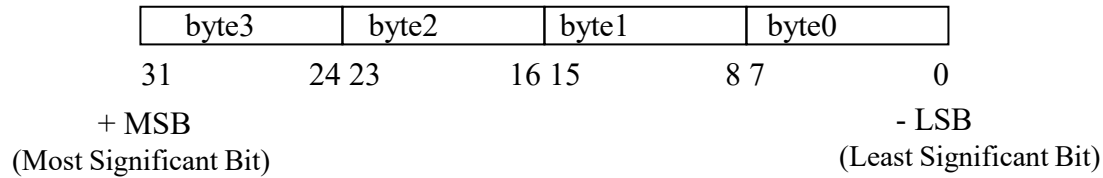
IBM
PowerPC™ (bi-endian)



Storage of words in memory

example

32-bit word



A	byte3
A+1	byte2
A+2	byte1
A+3	byte0

BigEndian

A	byte0
A+1	byte1
A+2	byte2
A+3	byte3

LittleEndian

The number $27_{(10)} = 11011_{(2)} = 00000000000000000000000000011011$

A	00000000
A+1	00000000
A+2	00000000
A+3	00011011

BigEndian

A	00011011
A+1	00000000
A+2	00000000
A+3	00000000

LittleEndian

Communication problems in computers with different architectures

The number $27_{(10)}$ is represented

00000000000000000000000000011011



A	00000000
A+1	00000000
A+2	00000000
A+3	00011011

**BigEndian
Computer**

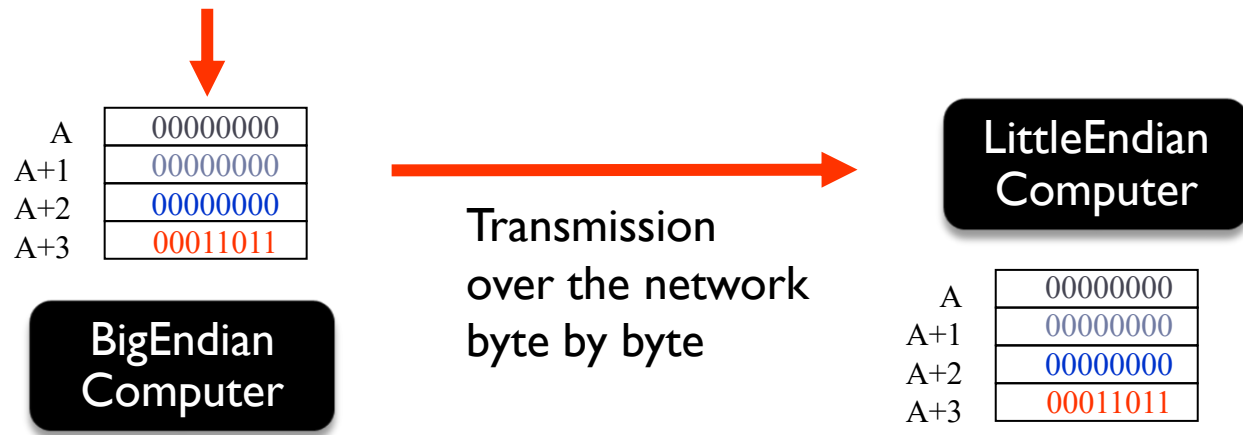
**LittleEndian
Computer**

A	
A+1	
A+2	
A+3	

Communication problems in computers with different architectures

The number $27_{(10)}$ is represented

00000000000000000000000000011011



Communication problems in computers with different architectures

The number $27_{(10)}$ is represented

00000000000000000000000000011011



A	00000000
A+1	00000000
A+2	00000000
A+3	00011011

**BigEndian
Computer**

Transmission
over the network
byte by byte

**LittleEndian
Computer**

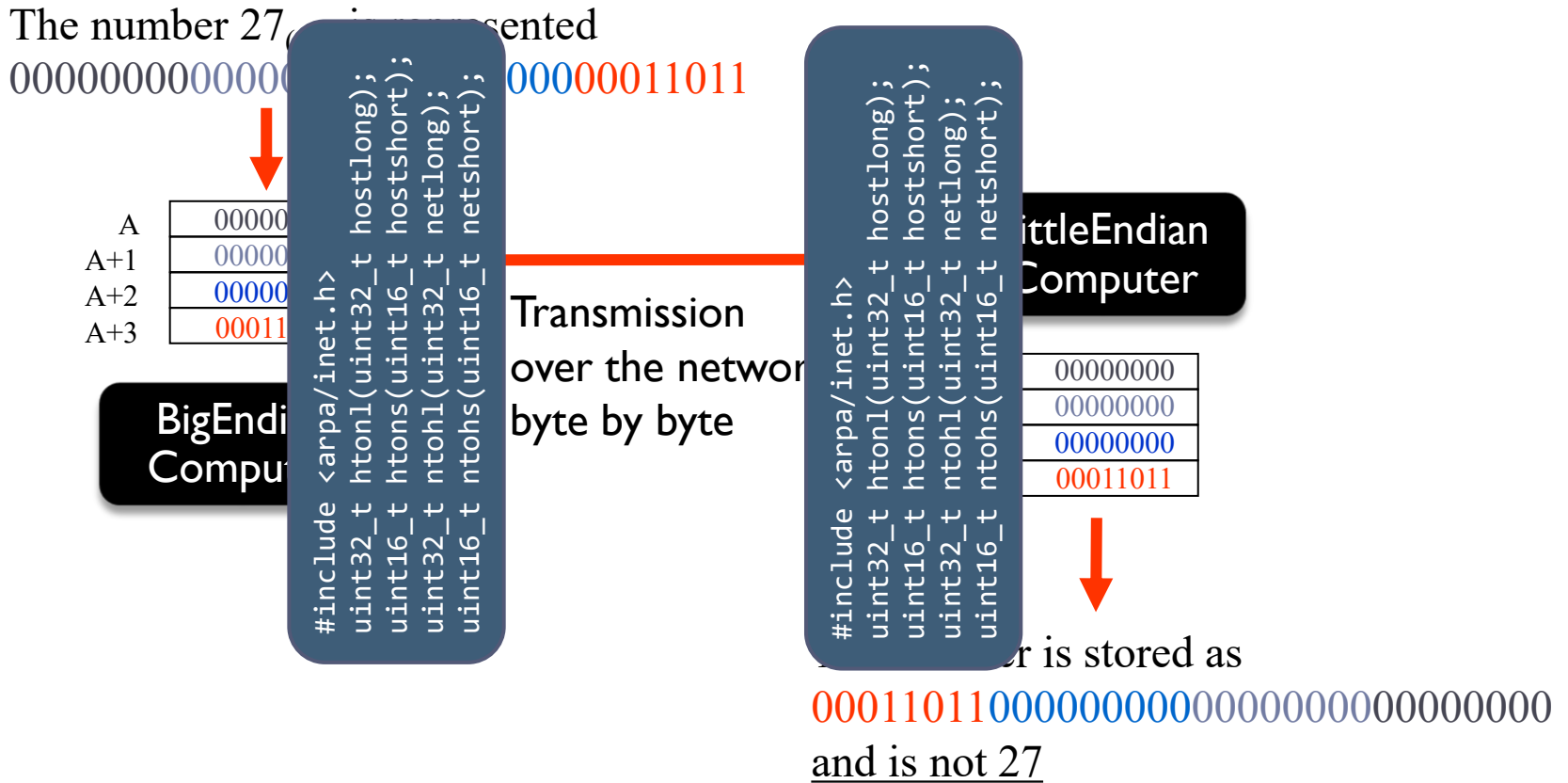
A	00000000
A+1	00000000
A+2	00000000
A+3	00011011



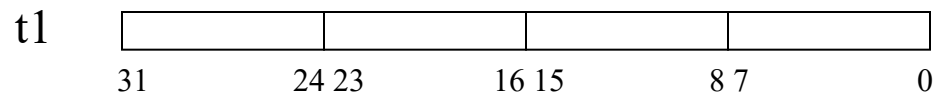
The number is stored as

00011011000000000000000000000000
and is not 27

Communication problems in computers with different architectures

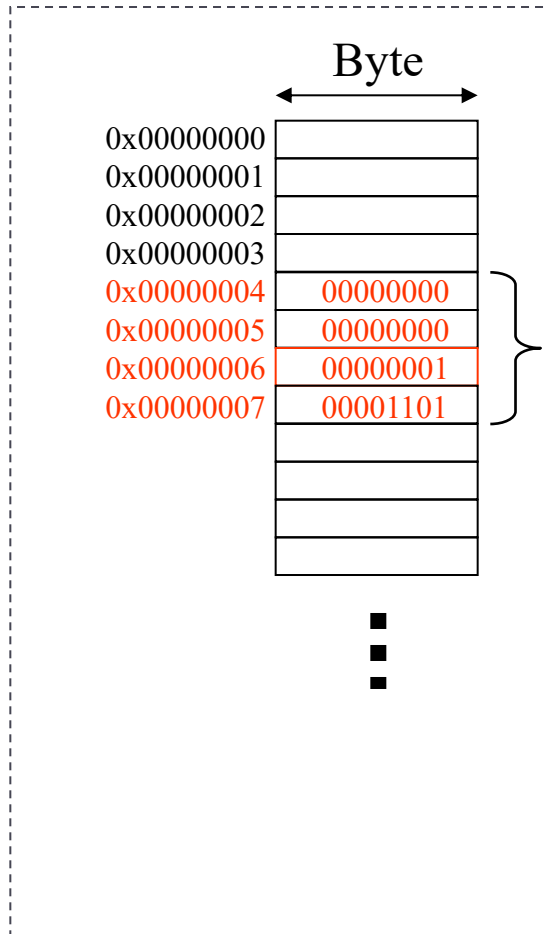


First byte address is specified: access to the stored word from address 0x00000004

[illegible]

Access to words

```
lw t1, 0x4(x0)
```

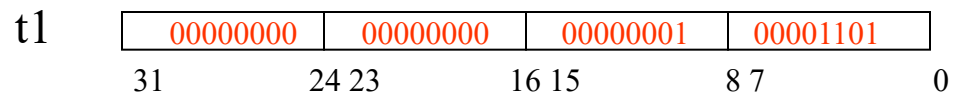


Address: 0x00000004 (000000.....00100)

[illegible]

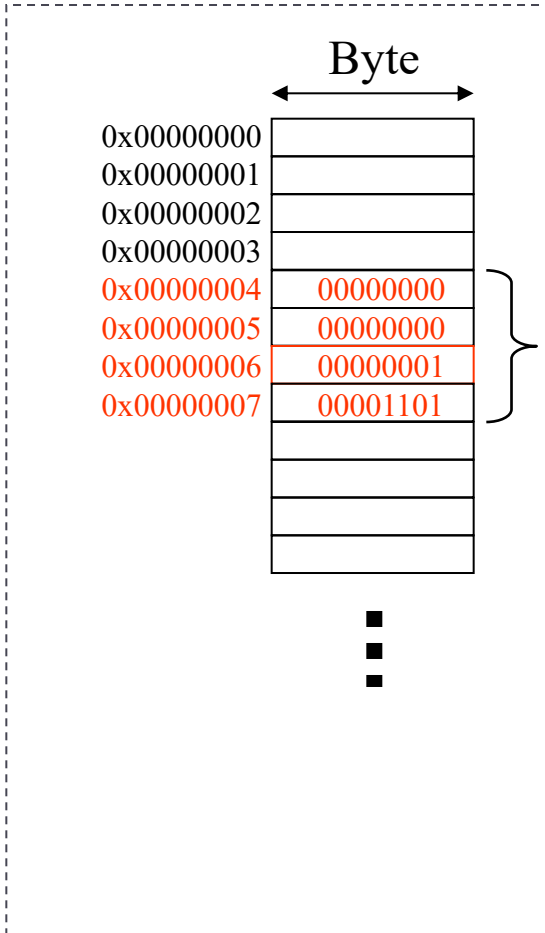
Computador Big Endian

The word is copied



Access to words

```
lw t1, 0x4(x0)
```



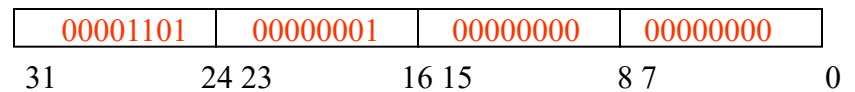
Address: 0x00000004 (000000.....00100)

[illegible]

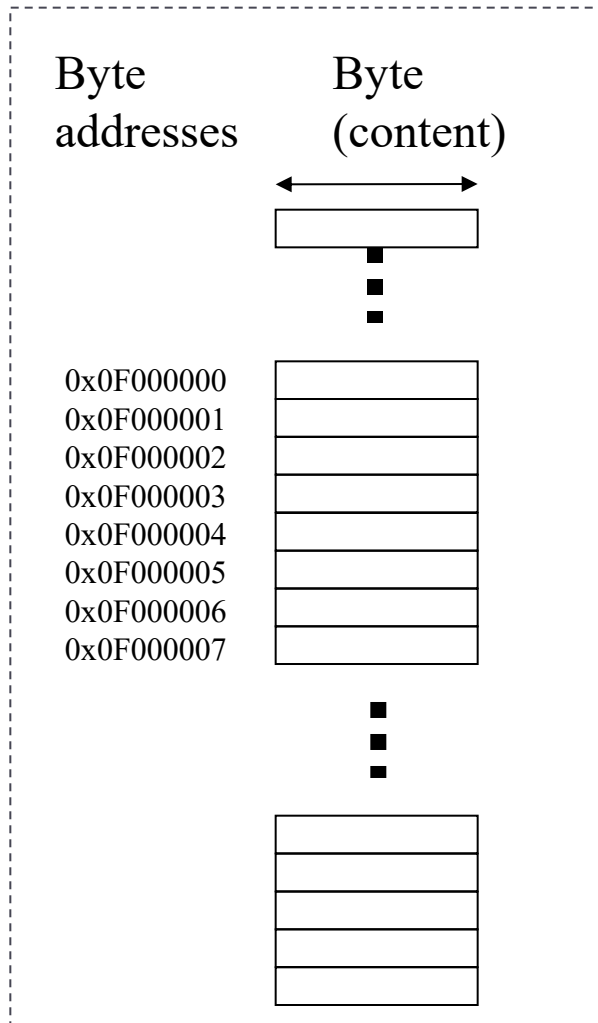
Computador Little Endian

The word is copied

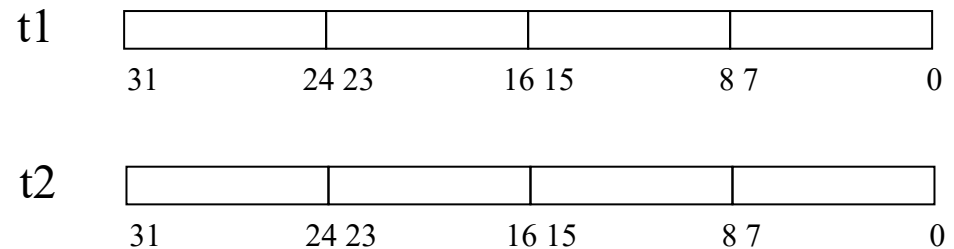
t1



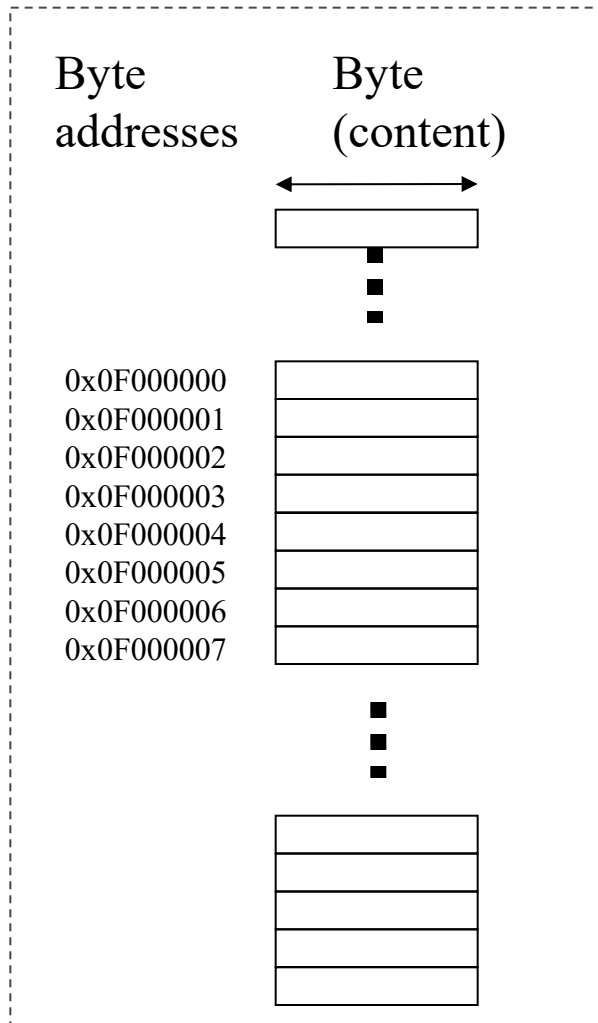
Writing to memory



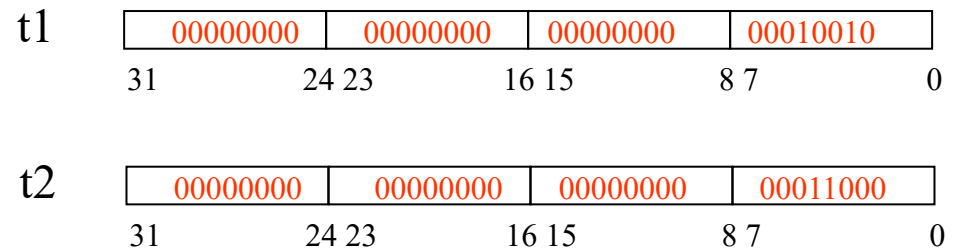
```
li t1, 18
li t2, 24
la t0, 0x0F000001
sb t1, 0(t0)
la t0, 0x0F000000
sw t1, 0(t0)
la t0, 0x0F000004
sw t2, 0(t0)
```



Writing to memory



```
li t1, 18
li t2, 24
la t0, 0x0F000001
sb t1, 0(t0)
la t0, 0x0F000000
sw t1, 0(t0)
la t0, 0x0F000004
sw t2, 0(t0)
```



Writing to memory

write one byte

```
li t1, 18
li t2, 24
la t0, 0x0F000001
sb t1, 0(t0)
la t0, 0x0F000000
sw t1, 0(t0)
la t0, 0x0F000004
sw t2, 0(t0)
```

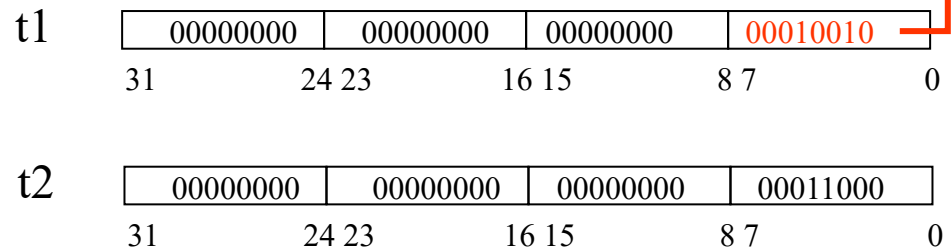
Byte
addresses

Byte
(content)

0x0F000000
0x0F000001
0x0F000002
0x0F000003
0x0F000004
0x0F000005
0x0F000006
0x0F000007

00010010

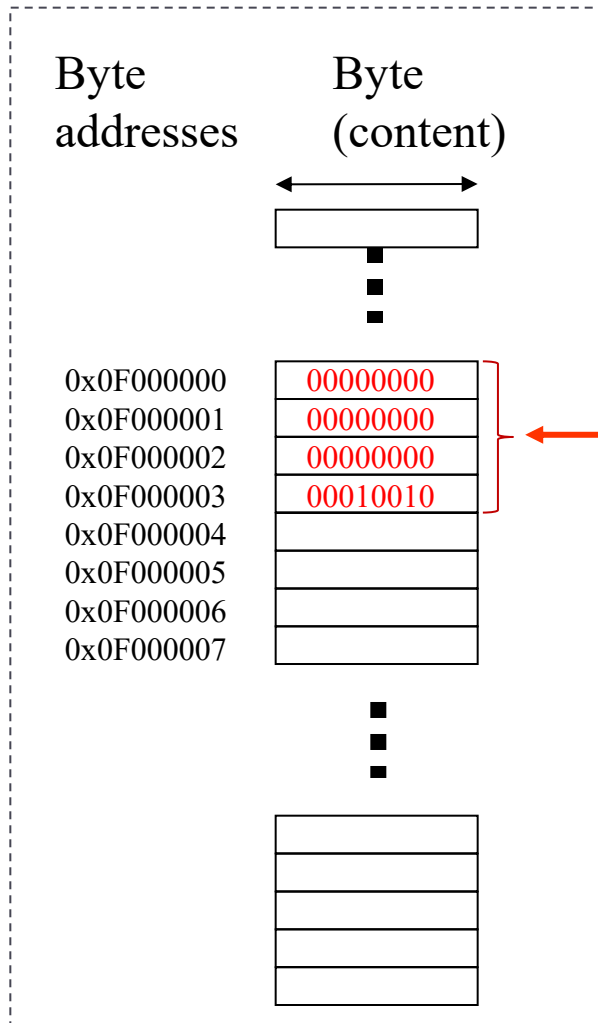
Writes the contents of the **least significant byte** of register t1 to memory



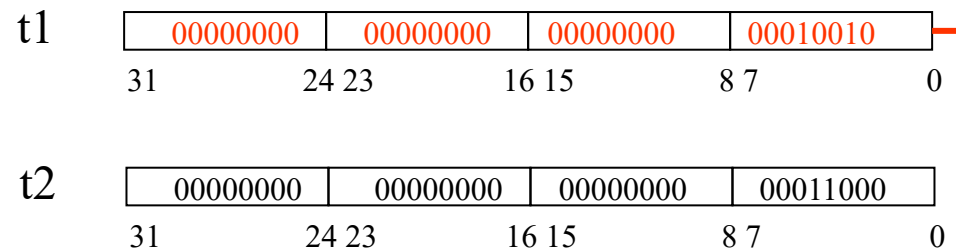
Writing to memory

write one word

```
li t1, 18
li t2, 24
la t0, 0x0F000001
sb t1, 0(t0)
la t0, 0x0F000000
sw t1, 0(t0)
la t0, 0x0F000004
sw t2, 0(t0)
```



Writes the content of a record to memory (the complete word)



Writing to memory

write one word

```
li t1, 18
li t2, 24
la t0, 0x0F000001
sb t1, 0(t0)
la t0, 0x0F000000
sw t1, 0(t0)
la t0, 0x0F000004
sw t2, 0(t0)
```

Writes the content of a record to
memory (the complete word)

Byte
addresses

Byte
(content)

0x0F000000
0x0F000001
0x0F000002
0x0F000003
0x0F000004
0x0F000005
0x0F000006
0x0F000007

00000000
00000000
00000000
00010010
00000000
00000000
00000000
00011000

t1

00000000	00000000	00000000	00010010
----------	----------	----------	----------

31 24 23 16 15 8 7 0

t2

00000000	00000000	00000000	00011000
----------	----------	----------	----------

31 24 23 16 15 8 7 0

Review: memory access

- ▶ In order to be executed, a program must be loaded together with its data in memory.
 - ▶ All instructions and data are stored in memory.
 - ▶ Therefore, everything (instructions and data) has a memory address.
- ▶ In a 32-bits computer such as RISC-V 32:
 - ▶ Registers have 32 bits
 - ▶ Memory can store bytes (8 bits)
 - ▶ memory → register: `lb/lbu rd, num(rs1)`
 - ▶ register → memory: `sb rd, num(rs1)`
 - ▶ Memory can store words (32 bits)
 - ▶ memory → register: `lw rd, num(rs1)`
 - ▶ Register → memory: `sw rd, num(rs1)`

`num(reg)`: represents the address obtained by summing `num` with the address stored in the register

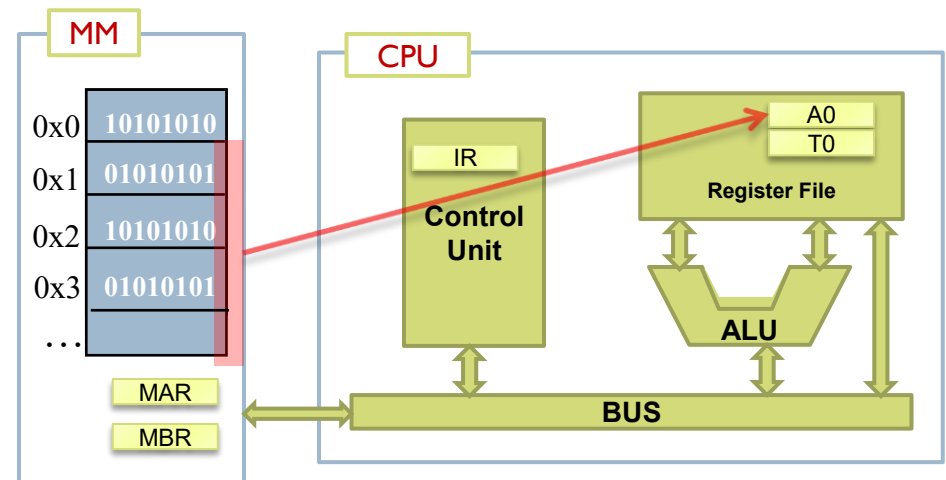
Examples of instructions

- ▶ `la t0, 0x0F000002`
 - ▶ The value `0x0F000002` is loaded at `t0`
- ▶ `lbu t0, label(x0)`
 - ▶ The byte at memory address `label` is loaded at `t0`.
- ▶ `lb t0, 0(t1)`
 - ▶ The byte in the memory location stored in `t1+0` is loaded in `t0`
- ▶ `la t0, 0x0F000000`
`sw t1, 0(t0)`
 - ▶ Copies the word stored in `t1` at address `0x0F000000`
- ▶ `la t0, 0x0F000000`
`sb t1, 0(t0)`
 - ▶ Copies the byte stored in `t1` (the least significant byte) to address `0x0F000000`

Data transfer alignment and access size

► Peculiarities:

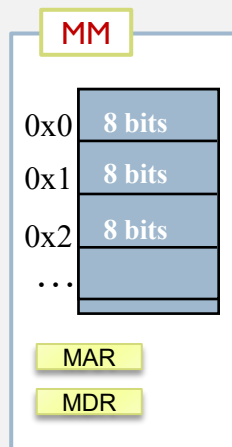
- Default access size
- Alignment of elements in memory



Word-level or byte-level addressing

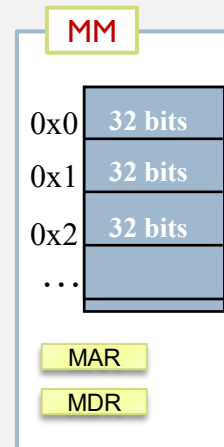
- ▶ The **main memory** is like a large one-dimensional vector of items.
- ▶ A **memory address** is the index of one item in the vector.
- ▶ There are two most-used **types of addressing**:

- ▶ **Byte addressing**



- ▶ Each memory element is 1 **byte**
- ▶ Transferring a **word** means transferring **4 bytes** (for 32-bit CPU)

- ▶ **Word addressing**



- ▶ Each memory element is a **word**
- ▶ **lb** means transferring one **word** and keeping one **byte**.

Data alignment

- ▶ In general:

- ▶ A data of K bytes is aligned when the address D used to access this data fulfills the condition:

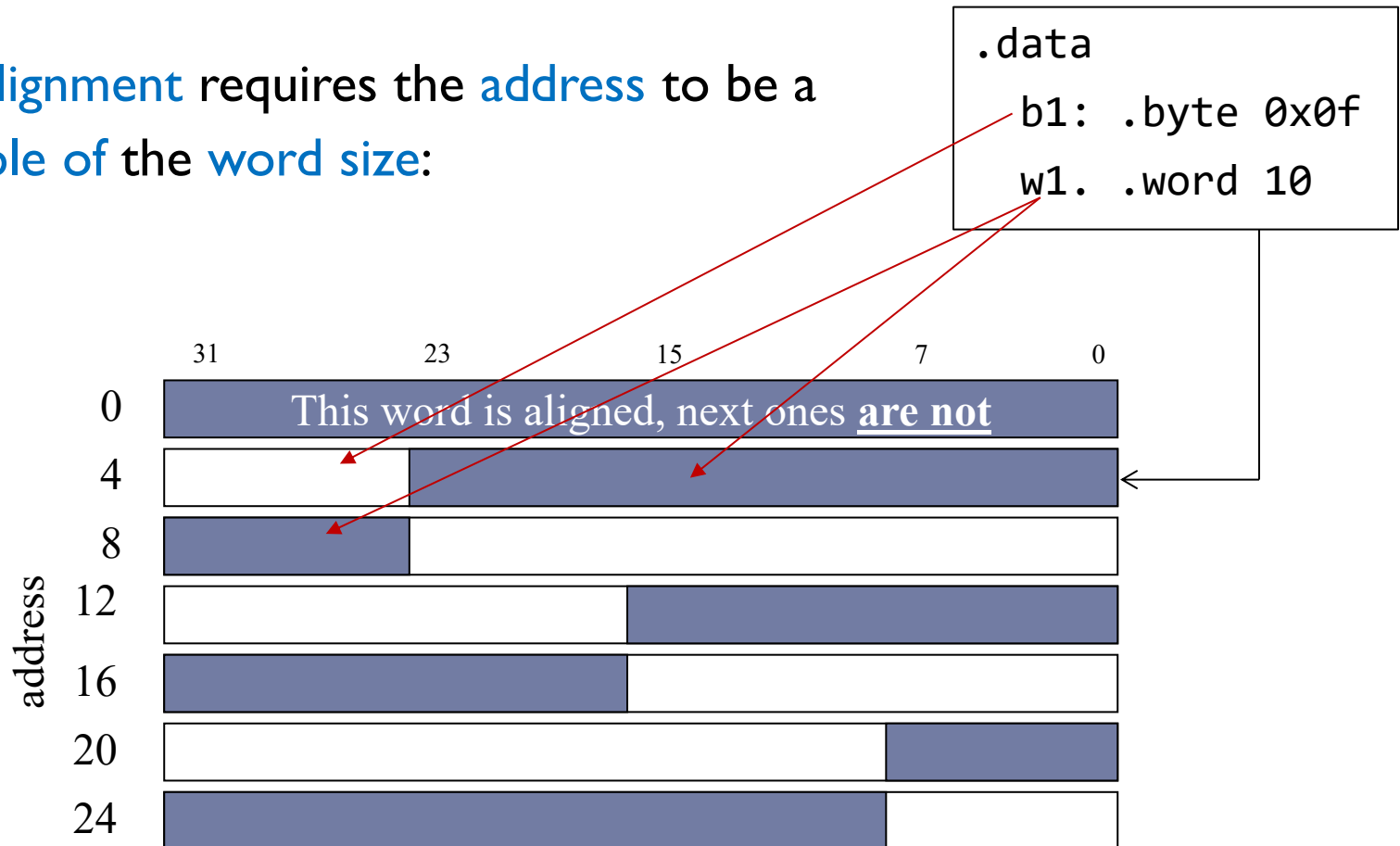
$$D \bmod K = 0$$

- ▶ Data alignment implies:

- ▶ Data of 2 bytes (.half)
are stored in even addresses (multiple of 2)
 - ▶ Data of 4 bytes (.word, .float)
are stored in addresses multiple of 4
 - ▶ Data of 8 bytes (.double)
are stored in addresses multiple of 8

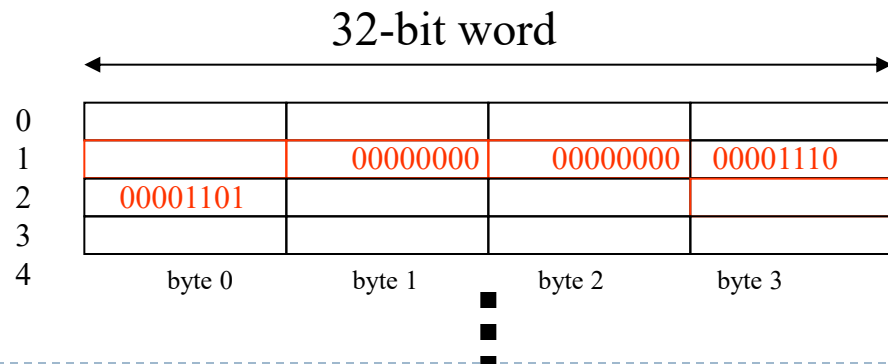
Data alignment: example

- The **alignment** requires the **address** to be a **multiple of the word size**:

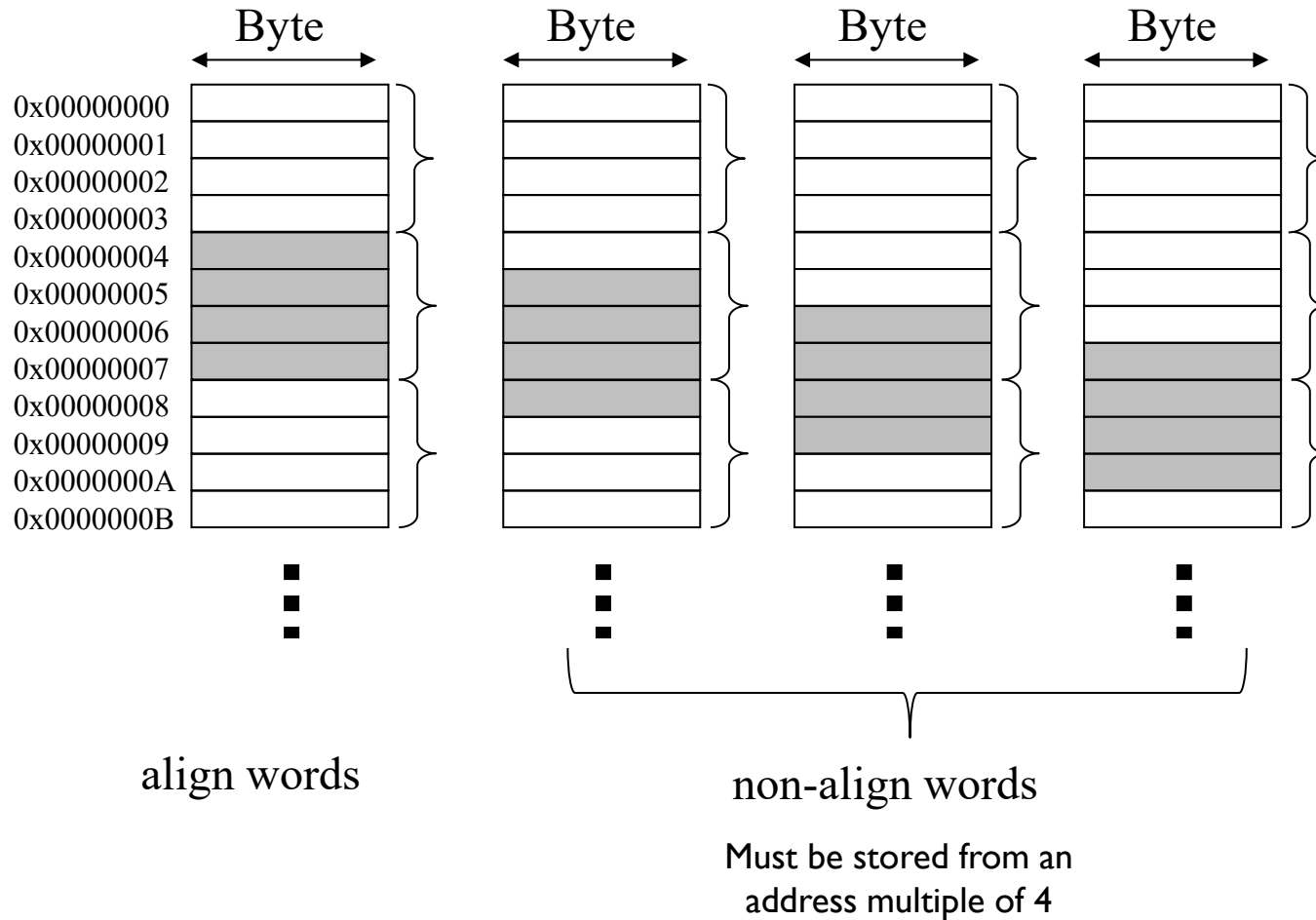


Data alignment

- ▶ Many computers does not allow the access to not aligned data:
 - ▶ Goal: reduce the number of memory accesses
 - ▶ Compilers assign addresses aligned to variables
- ▶ Some processors, such as Intel models, allow the access to not aligned data:
 - ▶ Non-aligned data needs several memory access



Non-aligned data



Data types in assembly

- ▶ Elemental data types
 - ▶ Booleans
 - ▶ Character
 - ▶ Integer
 - ▶ Float and double
- ▶ Compound data types
 - ▶ Vector: consecutives elements of the same type indexed by position
 - ▶ String
 - ▶ Matrix: two-dimensional indexed elements of the same type
 - ▶ Structs: consecutive elements of same/different types indexed by name

Elemental data types

Boolean

```
bool_t b1 = false;  
bool_t b2 = true;
```

```
...
```

```
main ()
```

```
{
```

```
    b1 = true ;
```

```
    ...
```

```
}
```

```
.data
```

```
b1: .byte 0          # 1 byte
```

```
b2: .byte 1
```

```
...
```

```
.text
```

```
main: la t0, b1
```

```
      li t1, 1
```

```
      sb t1, 0(t0)
```

```
...
```

Elemental data types

Character

```
char c1 ;  
char c2 = 'a';
```

```
...
```

```
main ()
```

```
{
```

```
    c1 = c2;
```

```
    ...
```

```
}
```

```
.data
```

```
c1: .zero 1 # 1 byte
```

```
c2: .byte 'a'
```

```
...
```

```
.text
```

```
main: la t0, c1
```

```
      la t1, c2
```

```
      lbu t2, 0(t1)
```

```
      sb t2, 0(t0)
```

```
...
```

Elemental data types

Integer

```
int  result ;  
int  op1 = 100 ;  
int  op2 = -10 ;
```

```
...
```

```
main ()
```

```
{  
    result = op1+op2;  
    ...  
}
```

```
.data  
  
.align 2  
resultado:  .zero 4  # 4 bytes  
op1:        .word 100  
op2:        .word -10  
...
```

```
.text
```

```
main:
```

```
    la  t0, op1  
    lw  t1, 0(t0)  
    la  t0, op2  
    lw  t2, 0(t0)  
    add t3, t1, t2  
    la  t0, result  
    sw  t3, 0(t0)
```

```
...
```

Elemental data types

Integer

global variable without initial value

```
int  result ;
int  op1 = 100 ;
int  op2 = -10 ;
```

...

global variable with initial value

```
main ()
```

```
{
    result = op1+op2;
```

...

```
}
```

```
.data
.align 2
resultado:  .zero 4  # 4 bytes
op1:        .word 100
op2:        .word -10
...
```

```
.text
```

```
main:
```

```
    la  t0, op1
    lw  t1, 0(t0)
    la  t0, op2
    lw  t2, 0(t0)
    add t3, t1, t2
    la  t0, result
    sw  t3, 0(t0)
```

...

Compound data types

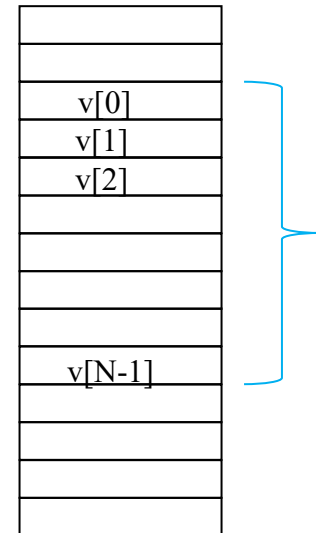
Arrays

- ▶ Collection of data items stored consecutively in memory
- ▶ The address of the j element can be computed as:

$$\text{init_address} + j * p$$

Where p is the size of each item

init_address



Compound data types

Arrays

```
int vec[5] ;
```

```
...
```

```
main ()
```

```
{
```

```
    vec[4] = 8;
```

```
}
```

```
.data
```

```
.align 2      # next item aligned to 4
```

```
vec: .zero 20 # 5 elem.*4 bytes
```

```
.text
```

```
main:
```

```
    la t1, vec
```

```
    li t2, 8
```

```
    sw t2, 12(t1)
```

```
    ...
```

Compound data types

Arrays

```
int vec[5] ;
```

```
...
```

```
main ()
```

```
{
```

```
    vec[4] = 8;
```

```
}
```

```
.data
```

```
.align 2      # next item aligned to 4
```

```
vec: .zero 20 # 5 elem.*4 bytes
```

```
.text
```

```
main:
```

```
    li  t2, 8
```

```
    li  t0, 16
```

```
    la  t1, vec
```

```
    add t3, t1, t0
```

```
    sw  t2, 0(t3)
```

```
    ...
```

Compound data types

Arrays

```
int vec[5] ;
```

```
...
```

```
main ()
```

```
{
```

```
    vec[4] = 8;
```

```
}
```

```
.data
```

```
.align 2 # next item aligned to 4  
vec: .zero 20 #5 elem.*4 bytes
```

```
.text
```

```
main:
```

```
    li t2, 8  
    li t1, 12  
    sw t2, vec(t1)  
    ...
```

Exercise

- ▶ Let V be an array of integer elements
 - ▶ V represents the initial address of the array
- ▶ What is the address of the $V[5]$ item?
- ▶ Which are the instructions to load in register $\$t0$ the value of $v[5]$?

Exercise (solution)

- ▶ Let V be an array of integer elements
 - ▶ V represents the initial address of the array
- ▶ What is the address of the $V[5]$ item?
 - ▶ $V + 5 * 4$
- ▶ Which are the instructions to load in register $\$t0$ the value of $v[5]$?
 - ▶ `li t1, 20`
 - ▶ `lw t0, v(t1)`

Compound data types

String

CREATOR

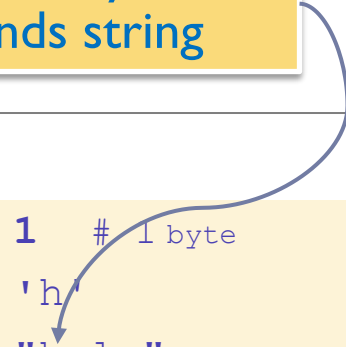
WepSIM

- Array of bytes
- '\0' ends string

```
char c1 ;  
char c2='h' ;  
char *ac1 = "hola" ;  
...
```

```
main ()  
{  
    printf("%s",ac1) ;  
    ...  
}
```

```
.data  
c1:  .zero    1    # 1 byte  
c2:  .byte    'h'  
ac1: .string  "hola"  
...
```



```
.text
```

```
main:  
    li a7, 4  
    la a0, ac1  
    ecall  
    ...
```

Exercise

```
// global variables
```

```
char v1;
```

```
int v2 ;
```

```
float v3 = 3.14 ;
```

```
char v4 = "ec" ;
```

```
int v5[] = { 20, 22 } ;
```

Exercise (solution)

```
// global variables
```

```
char v1;
```

```
int v2 ;
```

```
float v3 = 3.14 ;
```

```
char v4 = "ec" ;
```

```
int v5[] = { 20, 22 } ;
```

```
.data
```

```
v1: .byte 0
```

```
.align 2
```

```
v2: .zero 4
```

```
v3: .float 3.14
```

```
v4: .string "ec"
```

```
.align 2
```

```
v5: .word 20, 22
```

Exercise (solution)

v1:	0	0x0100
	?	0x0101
	?	0x0102
	?	0x0103
	?	
	?	
	?	
	?	
	?	
	?	
	?	
	?	
	?	
	?	
	?	
	?	
	?	
	...	

.data

v1: .byte 0

.align 2

v2: .zero 4

v3: .float 3.14

v4: .string "ec"

.align 2

v5: .word 20, 22

Exercise (solution)

v1:	0	0x0100
	?	0x0101
	?	0x0102
	?	0x0103
v2:	0	0x0104
	0	0x0105
	0	0x0106
	0	0x0107
v3:	(3.14)	0x0108
	(3.14)	0x0109
	(3.14)	0x010A
	(3.14)	0x010B
v4:	'e'	0x010C
	'c'	0x010D
	0	0x010E
		0x010F
v5:	(20)	0x0110
	(20)	0x0111
	(20)	0x0112
	(20)	

```
.data

v1: .byte 0
    .align 2
v2: .zero 4
v3: .float 3.14

v4: .string "ec"

    .align 2
v5: .word 20, 22
```

Compound data types

String length

```
char  c1 ;
char  c2  = 'h' ;
char *ac1 = "hola" ;
char *c;
...

main ()
{
    c = ac1; int l = 0;
    while (c[l] != NULL) {
        l++;
    }
    printf("%d", l);
    ...
}
```

Compound data types

String length

```
char  c1 ;
char  c2  = 'h' ;
char *ac1 = "hola" ;
char *c;
```

...

```
main ()
```

```
{
    c = ac1; int l = 0;
    while (c[l] != NULL) {
        l++;
    }
    printf("%d", l);
    ...
}
```

```
.data
c1:  .zero    1      # 1 byte
c2:  .byte    'h'
ac1: .string  "hola"
.align 2
c:   .zero    4      # pointer => address
```

...

```
.text
```

```
main:      la     t0, ac1
           li     a0, 0
           lbu    t1, 0(t0)
buc1:      beq    x0, t1, fin1
           addi   t0, t0, 1
           addi   a0, a0, 1
           lbu    t1, 0(t0)
           j      buc1

fin1:      li     a7, 1
           ecall
           ...
```

Arrays and strings

► Review (in general) :

► `lw t0, 4(s3) # t0 ← M[s3+4]`

► `sw t0, 4(s3) # M[s3+4] ← t0`

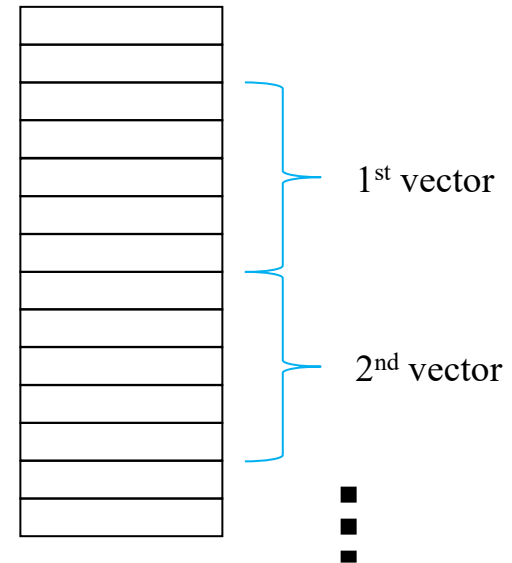
Compound data types

Matrix

- ▶ A matrix $m \times n$ consists of m vectors (m rows) of length n
- ▶ Usually stored by rows
- ▶ The element a_{ij} is stored in the address:

$$\text{init_address} + (i \cdot n + j) \times p$$

where p is the size of each item



Compound data types

Matrix

```
int vec[5] ;
int mat[2][3] = {{11,12,13},
                 {21,22,23}};
...
```

```
main ()
{
    m[0][1] = m[0][0] +
             m[1][0] ;
    ...
}
```

```
.data
.align 2 # next item aligned to 4
vec: .zero 20 #5 elem.*4 bytes
mat: .word 11, 12, 13
     .word 21, 22, 23
     ...
```

```
.text
main:
    li t0, 0
    lw t1, mat(t0)
    li t0, 12
    lw t2, mat(t0)
    add t3, t1, t2
    li t0, 4
    sw t3, mat(t0)
    ...
```

Tips

- ▶ Do not program directly in assembler
 - ▶ Better to **first do the design** in DFD, Java/C/Pascal...
 - ▶ Gradually translate the design to assembler.
- ▶ Sufficiently **comment** the code and data
 - ▶ By line or by group of lines
comment which part of the design implements.
- ▶ **Test** with enough test cases
 - ▶ Test that the final program works properly to the given specifications.

ARCOS Group

uc3m | Universidad **Carlos III** de Madrid

L3: Fundamentals of assembler programming (2) Computer Structure

Bachelor in Computer Science and Engineering

Bachelor in Applied Mathematics and Computing

Dual Bachelor in Computer Science and Engineering and Business Administration



Exercise

- ▶ Write a program that:
 - ▶ Loads the value -3.141516 into register f0
 - ▶ Allows you to retrieve the value of the exponent and the mantissa stored in register f0 (in IEEE 754 format)
 - ▶ Prints the sign
 - ▶ Prints the exponent
 - ▶ Prints the mantissa

Exercise (solution)

```
.data
    valor:      .float -3.141516
```

```
.text
```

```
main:
```

```
    la    t0, valor
    flw   ft0, 0(t0)
```

```
    # print
```

```
    fmv.s fa0, ft0
```

```
    li    a7, 2
```

```
    ecall
```

```
    li    a0, '\n' # intro
```

```
    li    a7, 11
```

```
    ecall
```

```
    # copy to cpu
```

```
    fmv.x.w    t0, ft0
```

```
li    t1, 0x80000000 # sign
and    a0, t0, t1
srli    a0, a0, 31
li    a7, 1
ecall
```

```
li    a0, '\n' # intro
li    a7, 11
ecall
```

```
li    t1, 0x7F800000 # exponent
and    a0, t0, t1
srli    a0, a0, 23
li    a7, 1
ecall
```

```
li    a0, '\n' # intro
li    a7, 11
ecall
```

```
li    t1, 0x007FFFFF # mantissa
and    a0, t0, t1
li    a7, 1
ecall
```

```
jr    ra
```